



US012410233B2

(12) **United States Patent**
Sung et al.

(10) **Patent No.: US 12,410,233 B2**
(45) **Date of Patent: *Sep. 9, 2025**

(54) **EXTRACELLULAR DOMAIN OF ALPHA SUBUNIT OF IgE Fc RECEPTOR, PHARMACEUTICAL COMPOSITION COMPRISING SAME AND METHOD FOR PRODUCING SAME**

2011/0256641 A1* 10/2011 Ling et al. G01N 33/6854
436/513
2012/0276096 A1 11/2012 Yang et al.
2018/0193452 A1* 7/2018 Ma C07K 16/2878

FOREIGN PATENT DOCUMENTS

(71) Applicant: **GI INNOVATION, INC.**, Seoul (KR)
(72) Inventors: **Young Chul Sung**, Seoul (KR);
Zungyoon Yang, Incheon (KR); **Myung Ho Jang**, Seoul (KR)
(73) Assignee: **GI Innovation, Inc.**, Seoul (KR)
(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 176 days.
This patent is subject to a terminal disclaimer.

CN	101611052	A	12/2009	
CN	105175553	A	12/2015	
JP	2007-528194	A	10/2007	
JP	2009-504569	A	2/2009	
KR	10-2017-0120579	A	10/2017	
RU	2500686	C2	12/2013	
TW	201636373	A	10/2016	
TW	201930343	A	8/2019	
WO	2005/017148	A1	2/2005	
WO	2007/005786	A2	1/2007	
WO	WO-2008028068	A2 *	3/2008	 C07K 14/70535
WO	2008116149	A2	9/2008	
WO	WO-2008147143	A2 *	12/2008	 A61P 1/00
WO	2009/080816	A1	7/2009	
WO	2016/108654	A1	7/2016	
WO	2016/133197	A1	8/2016	
WO	WO-2016200219	A1 *	12/2016	 A61K 38/00
WO	2019/135666	A1	7/2019	

(21) Appl. No.: **16/958,861**

(22) PCT Filed: **Jan. 8, 2019**

(86) PCT No.: **PCT/KR2019/000274**

§ 371 (c)(1),

(2) Date: **Jun. 29, 2020**

(87) PCT Pub. No.: **WO2019/135668**

PCT Pub. Date: **Jul. 11, 2019**

(65) **Prior Publication Data**

US 2021/0070833 A1 Mar. 11, 2021

(30) **Foreign Application Priority Data**

Jan. 8, 2018 (KR) 10-2018-0002248

(51) **Int. Cl.**

A23L 33/18 (2016.01)

A61K 38/00 (2006.01)

A61P 37/08 (2006.01)

C07K 14/735 (2006.01)

(52) **U.S. Cl.**

CPC **C07K 14/70535** (2013.01); **A23L 33/18** (2016.08); **A61P 37/08** (2018.01); **A61K 38/00** (2013.01)

(58) **Field of Classification Search**

CPC C07K 14/70535; C07K 2319/00; A23L 33/18; A61P 37/08; A61K 38/00; A23V 2200/304

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

12,128,076 B2* 10/2024 Jang A61K 38/1774
2008/0300188 A1 12/2008 Yang et al.
2011/0183363 A1* 7/2011 Fischer G01N 33/564
530/387.2

OTHER PUBLICATIONS

Putnam et al., Amino Acid Sequence of the First Constant Region Domain and the Hinge Region of the [delta] Heavy Chain of Human IgD, Oct. 1981, Proc. Natl. Acad. Sci. USA, vol. 78 No. 10, pp. 6168-6172 (Year: 1981).*

Schwarzbaum et al., Mapping of murine IgE epitopes involved in IgE-Fc[epsilon] receptor interactions, 1989, Eur. J. Immunol., vol. 19, pp. 1015-1023 (Year: 1989).*

Takahashi et al., The High Affinity IgE Receptor (FcεRI) as a Target for Anti-allergic Agents, 2005, Allergology International, vol. 54 No. 1, pp. 1-5 (Year: 2005).*

Rogers et al., Molecular Characterization of Immunoglobulin D in Mammals: Immunoglobulin Heavy Constant Delta Genes in Dogs, Chimpanzees, and Four Old World Monkey Species, 2006, Immunology, vol. 118, pp. 88-100 (Year: 2006).*

(Continued)

Primary Examiner — Vanessa L. Ford

Assistant Examiner — Danaya L. Middleton

(74) *Attorney, Agent, or Firm* — Sughrue Mion, PLLC

(57)

ABSTRACT

The present invention provides a polypeptide dimeric protein containing two monomers, each of which contains an extracellular domain (FcεRIa-ECD) of an alpha subunit of an IgE Fc receptor. The dimeric protein according to the present invention has advantages that an excellent binding ability to IgE is exhibited as compared with a conventional therapeutic agent containing an anti-IgE antibody, and less other side effects are exhibited due to lack of ADCC and CDC functions. Thus, the dimeric protein can be applied to a medical product for treating or preventing an IgE-mediated allergic disease.

12 Claims, 13 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited

OTHER PUBLICATIONS

Byrne et al., Sialic Acids: Carbohydrate Moieties That Influence The Biological and Physical Properties of Biopharmaceutical Proteins and Living Cells, Apr. 2007, Drug Discovery Today, vol. 12 Nos. 7/8, pp. 319-326 (Year: 2007).*

Saban et al., "Human Fc ϵ RI-IgG and humanized anti-IgE monoclonal antibody MaE11 block passive sensitization of human and rhesus monkey lung", Journal of Allergy and Clinical Immunology, 1994, vol. 94, No. 5, pp. 836-843 (8 pages total).

Haak-Frendscho et al., "Human IgE Receptor α -Chain IgG Chimera Blocks Passive Cutaneous Anaphylaxis Reaction In Vivo", The Journal of Immunology, 1993, vol. 151, pp. 351-358 (8 pages total).

Zoller et al., "Oligonucleotide-directed mutagenesis using M13-derived vectors: an efficient and general procedure for the production of point mutations in any fragment of DNA", Nucleic Acids Research, vol. 10, No. 20, 1982, p. 6487-6500 (14 pages).

S.C. Garman, et al., "Chain A, High Affinity Immunoglobulin Epsilon Receptor Alpha-Subunit", NCBI, Genbank Accession No. J189A, Oct. 10, 2012, 4 pages.

A.L. Flamar, et al., "Anti-human CD40 12E12 antibody HIV antigen fusion protein [synthetic construct]", NCBI, Genbank Accession No. AJD85779.1, Jan. 19, 2015, 3 pages.

M.B. White, et al., "immunoglobulin delta-chain, partial [*Homo sapiens*]", NCBI, Genbank Accession No. AAA52771.1, Aug. 1, 2016, 3 pages.

M.B. White, et al., "immunoglobulin delta-chain, partial [*Homo sapiens*]", NCBI, Genbank Accession No. AAA52770.1, Aug. 1, 2016, 3 pages.

Kyoko Takahashi, et al., "The High Affinity IgE Receptor (Fc ϵ RI) as a Target for Anti-allergic Agents", Allergy International, 2005, pp. 1-5, vol. 54, No. 1.

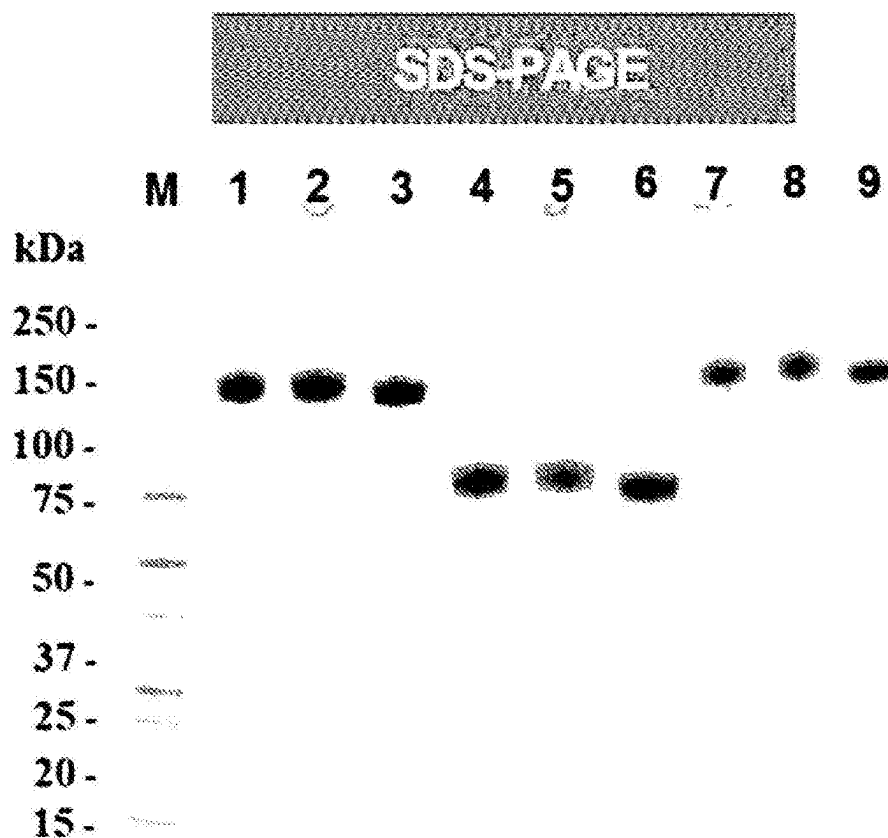
J.C. Venter, et al., "Fc fragment of IgE, high affinity I, receptor for; alpha polypeptide [*Homo sapiens*]", NCBI GenBank EAW52789.1, Mar. 23, 2015, 2 pages.

International Search Report for PCT/KR2019/000274 dated Apr. 15, 2019 (PCT/ISA/210).

Communication dated Feb. 18, 2023 issued by the State Intellectual Property Office of the P.R. China in application No. 2019800079175.

* cited by examiner

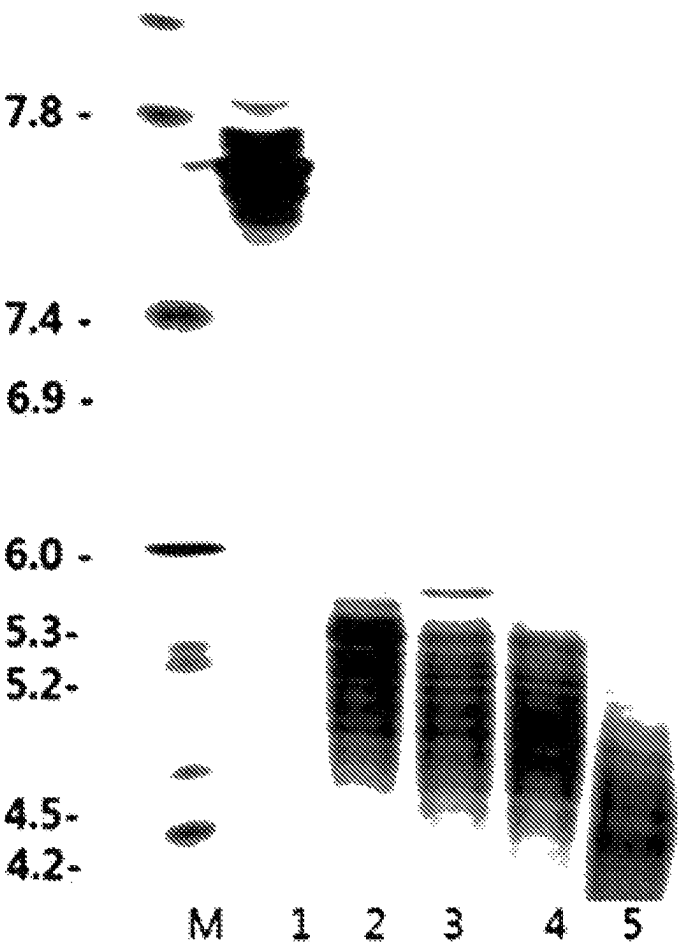
[FIG. 1A]



[FIG. 1B]

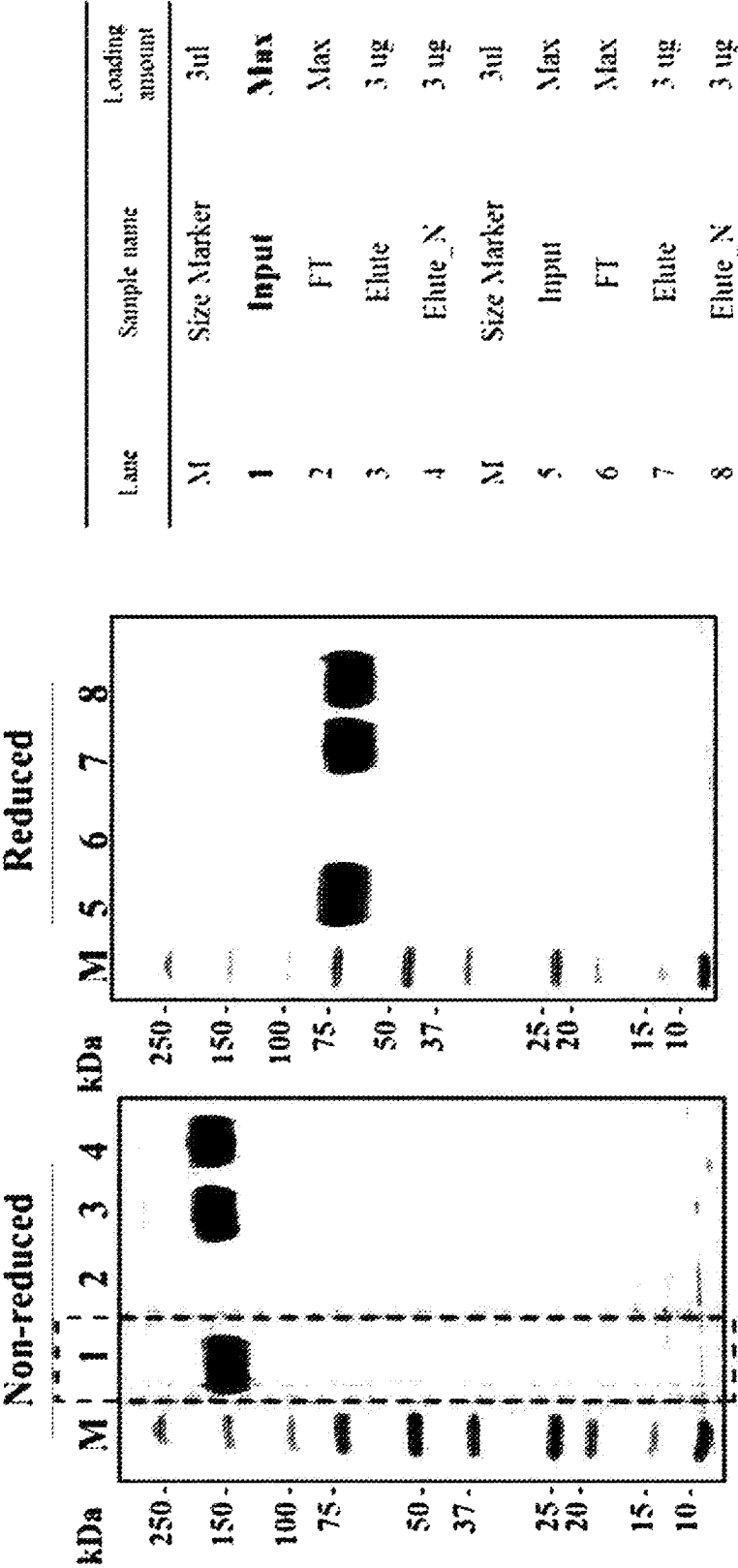
Lane #	Sample	Purification	Purity (SE-HPLC)	Sample condition	
M	Protein standard	One-step (Protein-A column) purification	-	-	-
1	FcεR1α ECD-Fc3		94.5%	-	Non-reducing
2	FcεR1α ECD-Fc3+2,6 ST		93.7%		
3	FcεR1α ECD-Fc2+2,6 ST		93.2%		
4	FcεR1α ECD-Fc3		94.5%	-	Reducing
5	FcεR1α ECD-Fc3+2,6 ST		93.7%		
6	FcεR1α ECD-Fc2+2,6 ST		93.2%		
7	FcεR1α ECD-Fc3		94.5%	Freezing/ Thawing test	Non-reducing
8	FcεR1α ECD-Fc3+2,6 ST		93.7%		
9	FcεR1α ECD-Fc3+2,6 ST		93.2%		

[FIG. 1C]

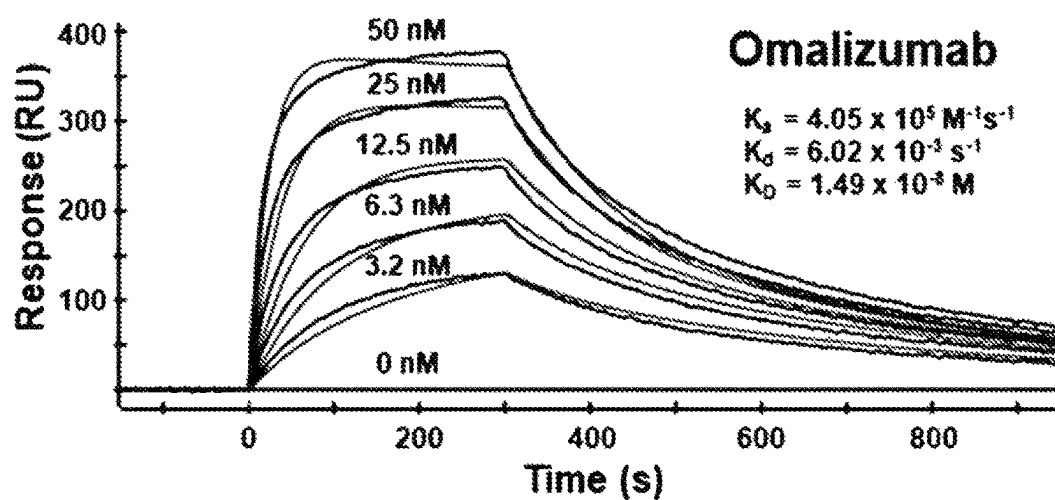


Lane	Loading sample	Theoretical pI
M	Marker (SERVA pH3-10, 5uL)	
1	Xolair	7.03
2	FcεRIα ECD-FC2	5.62
3	FcεRIα ECD-FC2+2,6 ST	5.62
4	FcεRIα ECD-FC3	5.63
5	FcεRIα ECD-FC3+2,6 ST	5.63

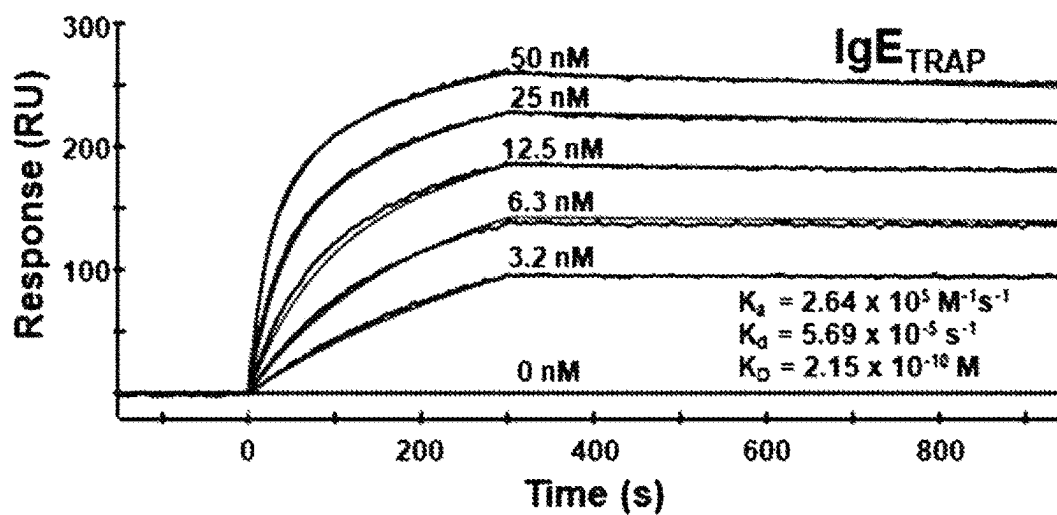
[FIG. 2]



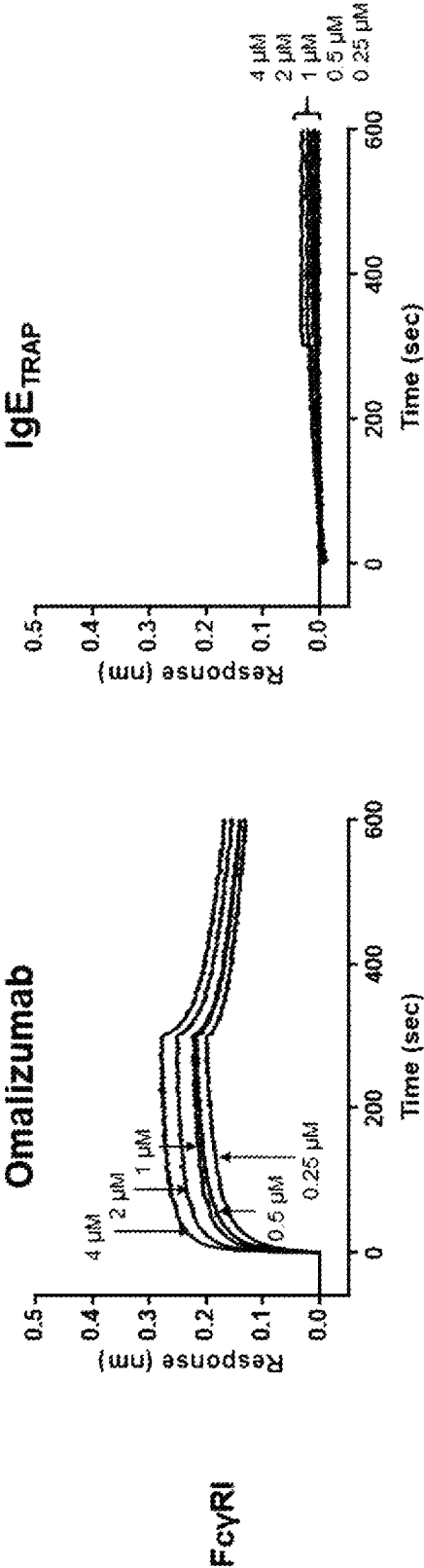
[FIG. 3]



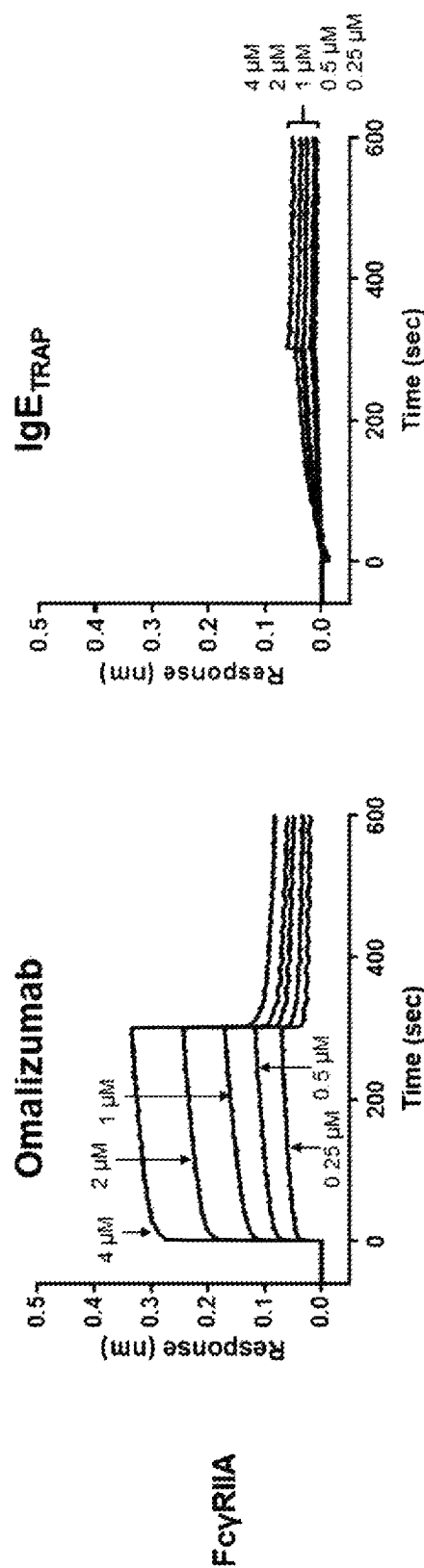
[FIG. 4]



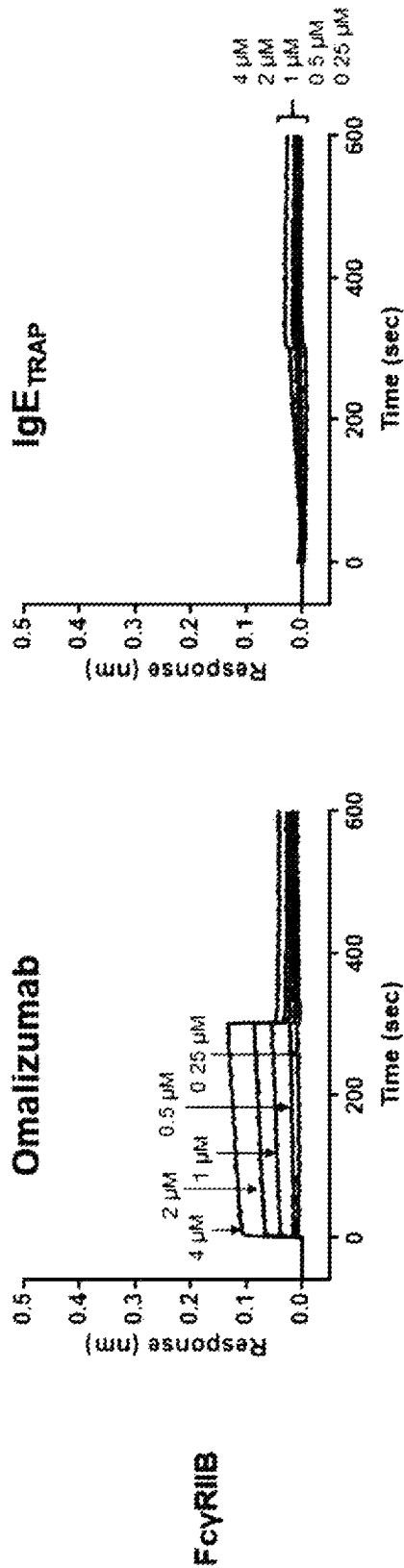
[FIG. 5A]



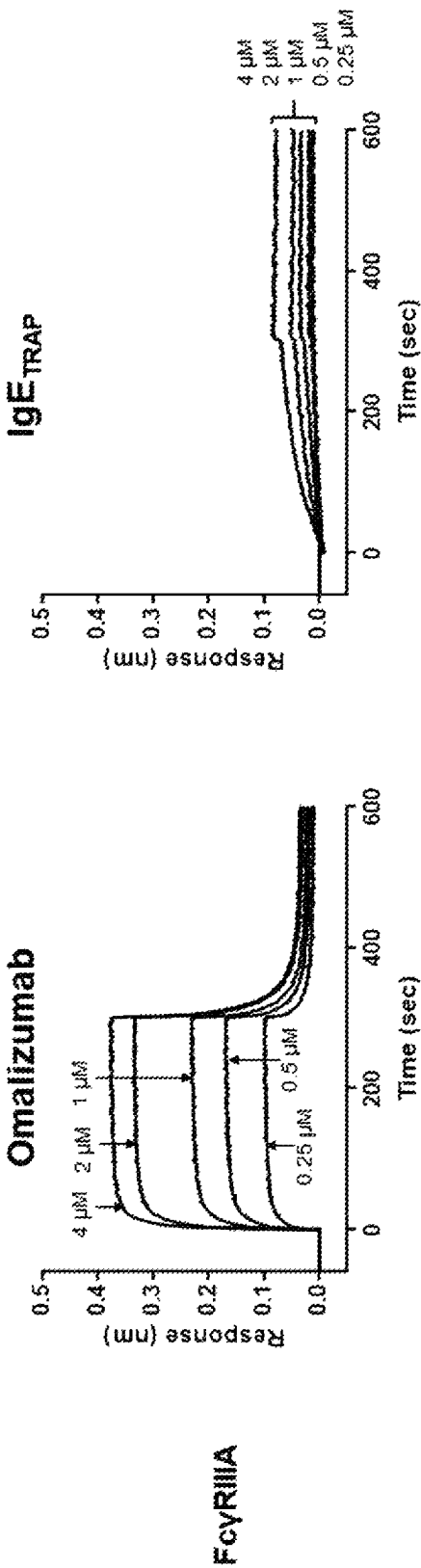
[FIG. 5B]



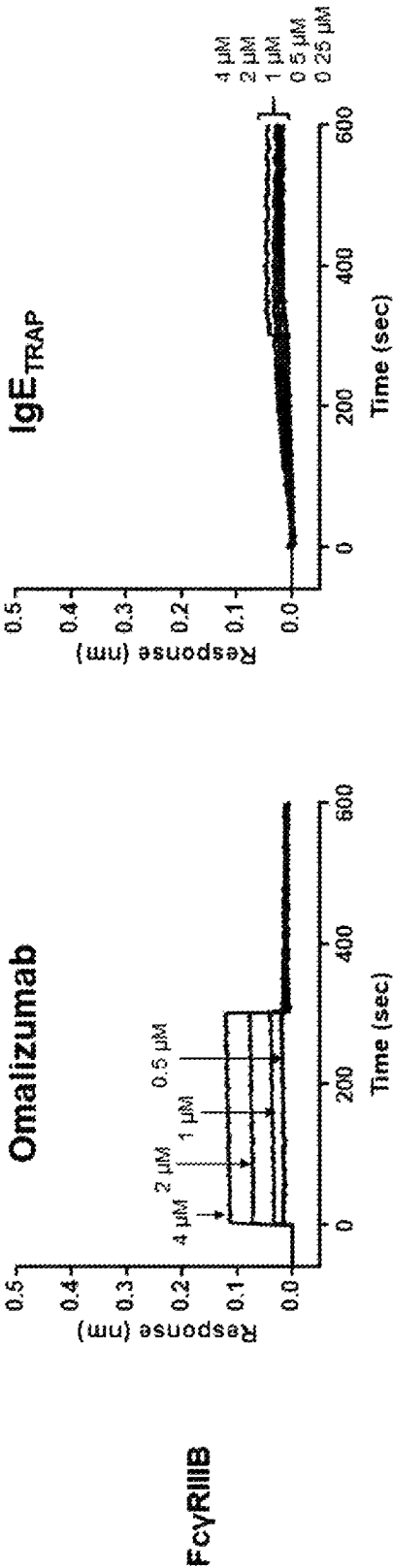
[FIG. 5C]



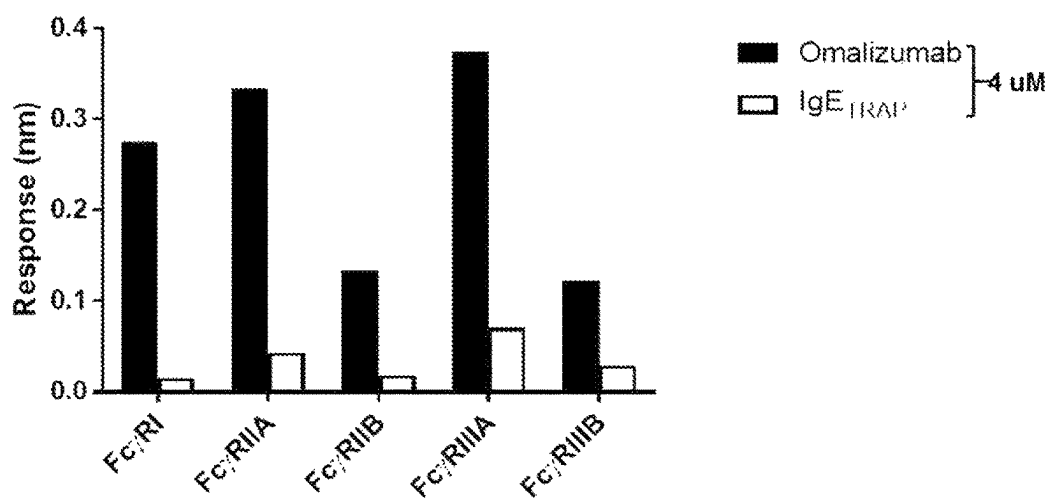
[FIG. 5D]



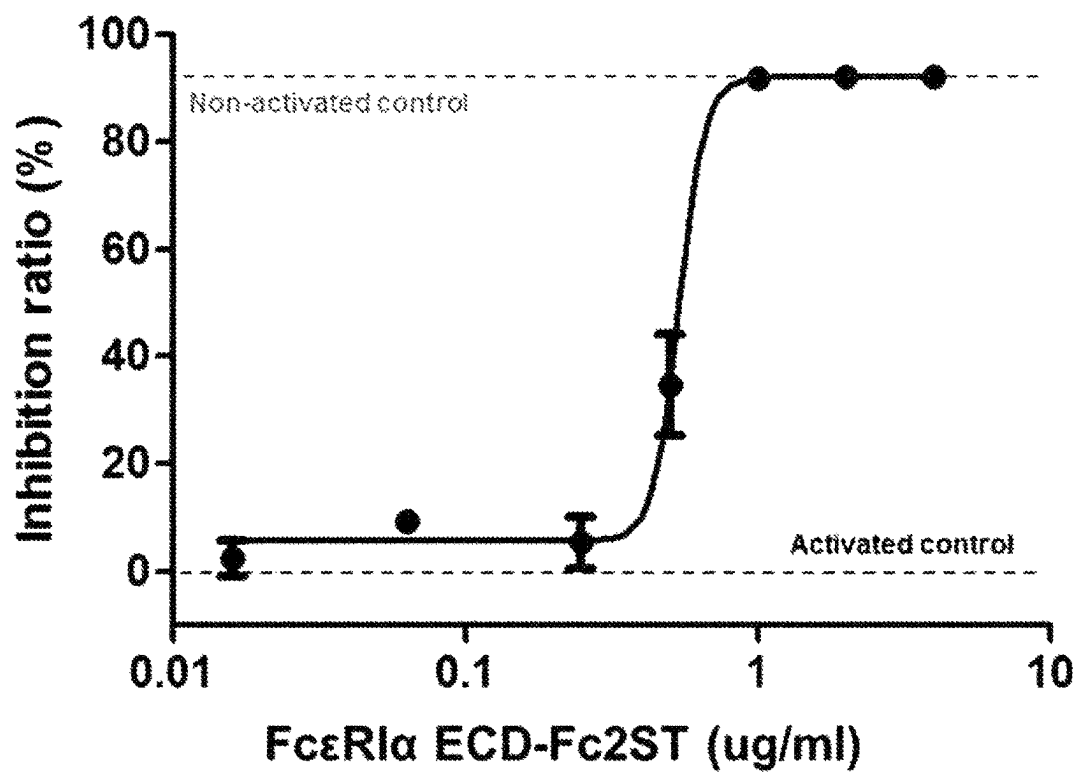
[FIG. 5E]



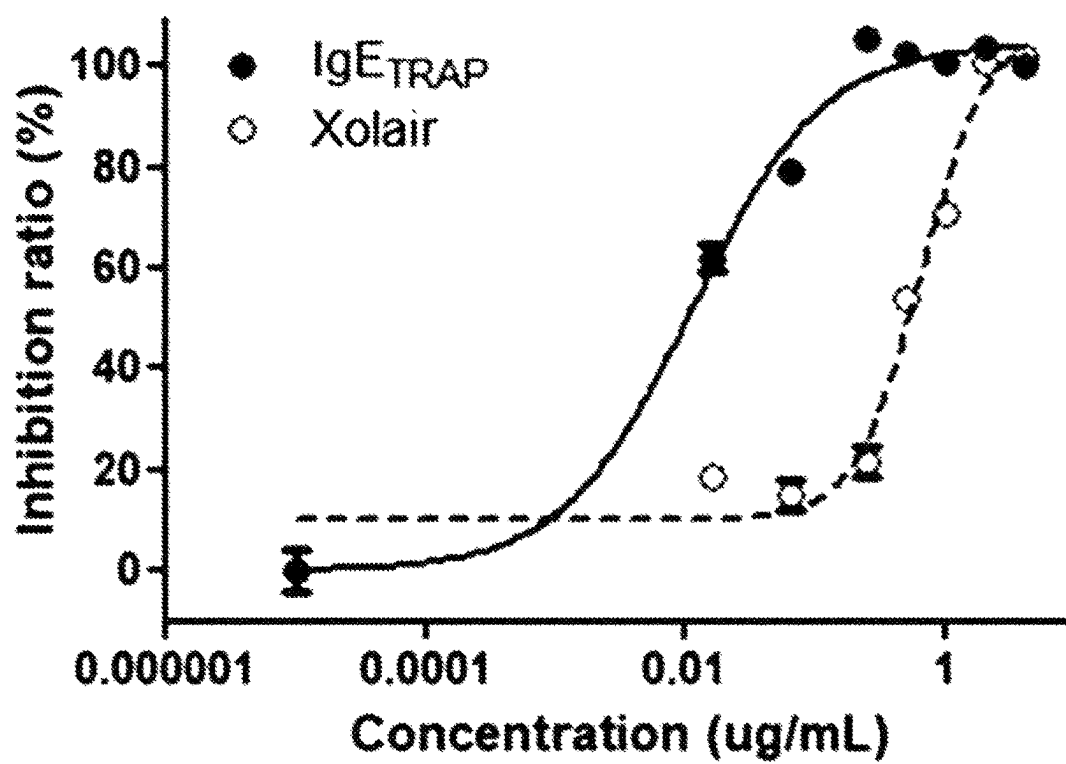
[FIG. 6]



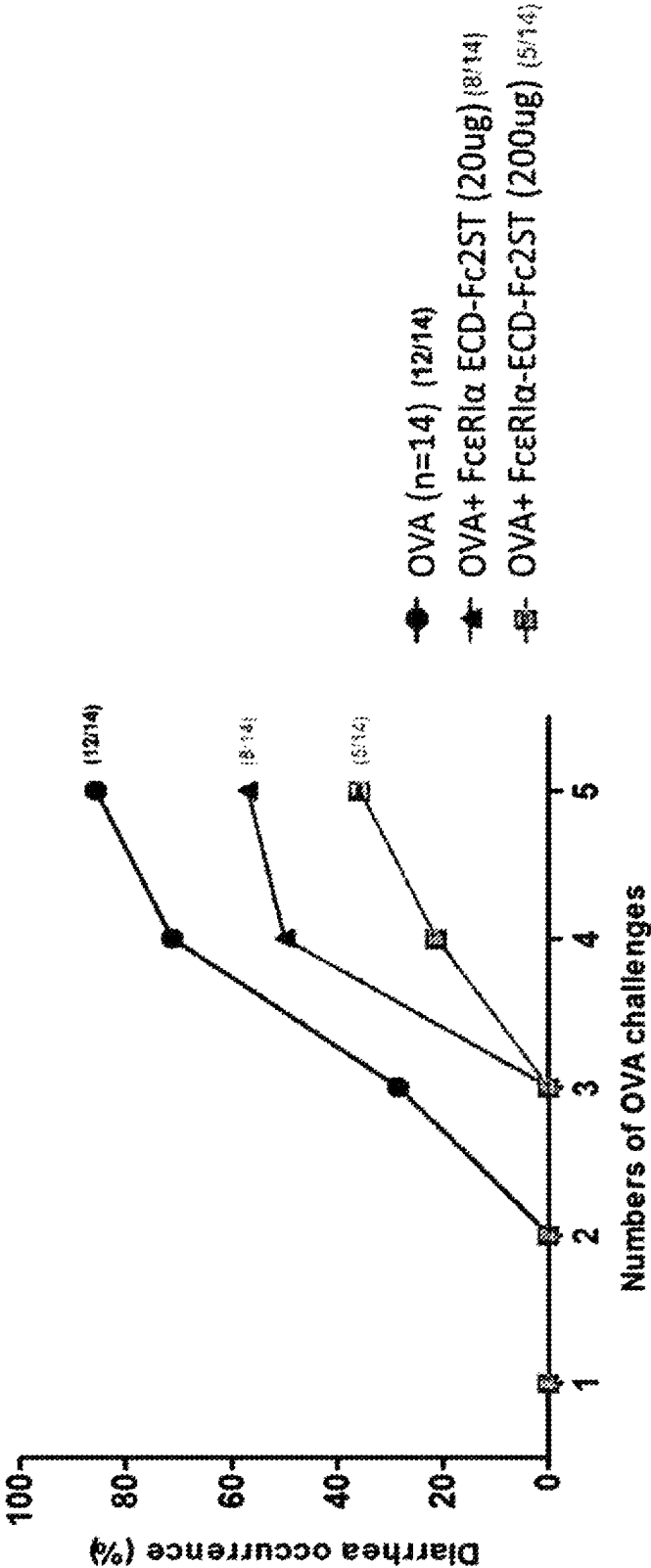
[FIG. 7]



[FIG. 8]



[FIG. 9]



1

**EXTRACELLULAR DOMAIN OF ALPHA
SUBUNIT OF IgE Fc RECEPTOR,
PHARMACEUTICAL COMPOSITION
COMPRISING SAME AND METHOD FOR
PRODUCING SAME**

**CROSS REFERENCE TO RELATED
APPLICATIONS**

This application is a National Stage of International Application No. PCT/KR2019/000274 filed Jan. 8, 2019, claiming priority based on Korean Patent Application No. 10-2018-0002248 filed Jan. 8, 2018.

**INCORPORATION BY REFERENCE OF
SEQUENCE LISTING**

The instant application contains a Sequence Listing which has been filed electronically in txt format and is hereby incorporated by reference in its entirety. Said txt copy, created on Jul. 1, 2020, is named Q255662SubstituteSequenceListingasfiled932020.txt and is 35.6 KB in size.

TECHNICAL FIELD

The present invention relates to modified IgE Fc receptors and uses thereof.

BACKGROUND ART

Allergic diseases, such as allergic rhinitis, atopic dermatitis, and food allergy, including asthma are spreading at a high rate in industrialized and westernized modern societies, and development of anaphylaxis, a severe allergic disease, is also increasing. These chronic immune diseases severely impair individuals' quality of life and socioeconomic costs are soaring accordingly. Thus, there is a desperate need for measures to overcome such diseases.

Most allergic diseases are caused by an excessive immune response of immunoglobulin E (IgE). IgE is an antibody that is present in serum at a very low concentration under a normal condition. IgE is also usually produced by innocuous antigens. There is a case where the number of IgE is increased without any particular stimulus. Such a case may lead to allergic diseases. The abnormally increased number of IgE can bind to high-affinity IgE Fc receptors (FcεR1s) which are expressed on the surface of mast cells, basophils, and the like. Such binding causes mast cells or basophils to release chemical mediators such as histamine, leukotriene, prostaglandin, bradykinin, and platelet-activating factors. Release of these chemical mediators results in allergic symptoms. In particular, allergic diseases may exhibit worsened symptoms due to the binding between IgE and FcεRI. FcεRI-expressing cells are known to increase in allergic patients.

Currently, various methods, such as allergen avoidance, administration of anti-allergic drugs, modulation of IgE synthesis in the body, and development of anti-IgE antibodies, have been proposed to treat allergic diseases. However, therapeutic methods known so far have many drawbacks, such as inability to cure an underlying cause of allergy, insufficient drug efficacy, and occurrence of serious side effects.

In addition, immunoglobulin compositions capable of binding to IgE and FcγRIIb with high affinity and inhibiting cells expressing IgE have been studied (KR10-1783272B1).

2

Such compositions have been reported to be useful for treating IgE-mediated disorders including allergy and asthma. In addition, omalizumab (trade name: XOLAIR®), which targets an Fc portion of an IgE antibody, has been developed and used as a therapeutic agent for intractable severe asthma and intractable urticaria.

However, a high-dose administration of omalizumab to maintain therapeutic effects leads to a high cost burden, and side effects such as angioedema and anaphylactic reaction (The Journal of Clinical Investigation Volume 99, Number 5, March 1997, 915-925). Besides, from the post-marketing results, allergic granulomatous vasculitis and idiopathic severe thrombocytopenia have been reported. Accordingly, there is a growing need for development of therapeutic agents capable of effectively treating allergic diseases without side effects.

Technical Problem

An object of the present invention is to provide a polypeptide dimeric protein for treating an IgE-mediated allergic disease. Another object of the present invention is to provide a nucleic acid molecule encoding the protein, an expression vector containing the nucleic acid molecule, and a host cell containing the expression vector. Yet another object of the present invention is to provide a method for preparing the polypeptide dimer.

Solution to Problem

In order to achieve the above objects, there is provided a polypeptide dimer, comprising two monomers, each of which contains an extracellular domain of an alpha subunit of an IgE Fc receptor. The monomer contains a modified Fc region, and the modified Fc region and the extracellular domain of the alpha subunit of the IgE Fc receptor are linked via a hinge of an IgD antibody. In another aspect, there is provided a pharmaceutical composition for treating or preventing an allergic disease, comprising the polypeptide dimer as an active ingredient.

Advantageous Effects of Invention

The polypeptide dimeric protein according to the present invention not only has excellent safety and persistence in the body as compared with conventionally used anti-IgE antibodies, but also binds to IgE very strongly due to having a binding capacity to IgE which is 70-fold higher than the conventionally used anti-IgE antibody, omalizumab, which allows an extended administration cycle. In addition, the polypeptide dimeric protein according to the present invention is a substance obtained by applying a modified Fc, which has IgE alone as a single target and does not bind to an Fc gamma receptor, and thus lacks antibody dependent cellular cytotoxicity (ADCC) and complement dependent cytotoxicity (CDC) functions.

Therefore, unlike conventional anti-IgE antibodies containing an IgG1 Fc region, the polypeptide dimeric protein does not bind to an Fc gamma receptor, and thus can inhibit release of mediators caused by being bound to the Fc gamma receptor on the surface of mast cells, so that severe side effects such as occurrence of anaphylaxis which can be caused by binding between IgG1 and Fc gamma receptor III on mast cells can be minimized. Therefore, the polypeptide dimeric protein according to the present invention can be

utilized as a new pharmaceutical composition which can replace therapeutic agents containing a conventional anti-IgE antibody.

BRIEF DESCRIPTION OF DRAWINGS

FIGS. 1A-1C show results of SDS-PAGE and gel isoelectric focusing (IEF) for proteins produced in each cell line. Here, it can be seen that a truncated form is not generated at both reducing and non-reducing conditions, and that a content of acidic proteins is increased due to an increase in sialic acid content caused by introduction of a sialic acid transferase gene.

FIG. 2 illustrates SDS-PAGE results for non-reduced and reduced forms of the polypeptide dimeric protein according to an embodiment of the present invention. In particular, it can be seen that the polypeptide dimer has high purity even in culture supernatant which corresponds to Input.

FIG. 3 illustrates a graph showing a binding ability of omalizumab to IgE. The graph shows results obtained by immobilizing omalizumab and analyzing a binding ability thereof depending on IgE concentrations treated.

FIG. 4 illustrates a graph showing a binding ability, to IgE, of the polypeptide dimeric protein according to an embodiment of the present invention. The graph shows results obtained by immobilizing the dimeric protein and analyzing a binding ability thereof depending on IgE concentrations treated.

FIGS. 5A-5E illustrate results obtained by identifying interactions of the polypeptide dimeric protein (IgE_{TRAP}), an embodiment of the present invention, and omalizumab with IgG receptors FcγRI (FIG. 5A), FcγRIIA (FIG. 5B), FcγRIIB (FIG. 5C), FcγRIIIA (FIG. 5D), and FcγRIIIB (FIG. 5E) by bio-layer interferometry (BLI) assay.

FIG. 6 illustrates a graph obtained by quantifying a binding capacity between IgE_{TRAP} and IgG receptors, and between omalizumab and IgG receptors.

FIG. 7 illustrates a graph showing an inhibitory ability, on activity of mouse-derived mast cells, of the polypeptide dimeric protein (IgE_{TRAP}) according to an embodiment of the present invention depending on concentrations thereof.

FIG. 8 illustrates a graph showing a comparison between inhibitory abilities, on activity of human FcεRI-expressing mouse-derived mast cells, of the polypeptide dimeric protein (IgE_{TRAP}) according to an embodiment of the present invention and XOLAIR® (omalizumab) depending on concentrations thereof.

FIG. 9 illustrates a graph showing administration effects of the polypeptide dimeric protein according to an embodiment of the present invention in a food allergy model.

DETAILED DESCRIPTION OF INVENTION

The present invention relates to a polypeptide dimer, comprising two monomers, each of which contains an extracellular domain (FcεRIα-ECD) of an alpha subunit of an IgE Fc receptor, in which the monomer contains a modified Fc region, and the modified Fc region and the FcεRIα-ECD are linked via a hinge of an IgD antibody.

As used herein, the term "IgE" means an antibody protein known as immunoglobulin E. IgE has an affinity to mast cells, blood basophils, or the like. In addition, reaction between an IgE antibody and an antigen (allergen) corresponding thereto causes an inflammatory reaction. In addition, IgE is known to be an antibody that causes anaphylaxis which occurs due to sudden secretion of mast cells or basophils.

As used herein, the term "IgE Fc receptor" is also referred to as Fcε receptor and binds to an Fc portion of IgE. There are two types for the receptor. The receptor having high affinity to IgE Fc is called Fcε receptor I (FcεRI). The receptor having low affinity to IgE Fc is called Fcε receptor II (FcεRII). FcεRI is expressed in mast cells and basophils. In a case where IgE antibodies bound to FcεRI are cross-linked by polyvalent antigens, degranulation occurs in mast cells or basophils, thereby releasing various chemical transmitter substances including histamine. This release leads to an immediate allergic reaction.

The FcεRI is a membrane protein composed of one α chain, one β chain, and two γ chains linked by a disulfide bond. Among these chains, a portion to which IgE binds is the α chain (FcεRIα). FcεRIα has a size of about 60 kDa, and is composed of a hydrophobic domain existing inside the cell membrane and a hydrophilic domain existing outside the cell membrane. In particular, IgE binds to an extracellular domain of the α chain.

Specifically, the alpha subunit of the IgE Fc receptor may have the amino acid sequence set forth in NP_001992.1. In addition, the extracellular domain (FcεRIα-ECD) of the alpha subunit of the IgE Fc receptor may have the amino acid sequence of SEQ ID NO: 1. In the present specification, the extracellular domain of the alpha subunit of the IgE Fc receptor may be a fragment or variant of the extracellular domain of the alpha subunit of the IgE Fc receptor, as long as the fragment or variant is capable of binding to IgE.

The variant may be prepared through a method of substituting, deleting, or adding one or more proteins in the wild-type FcεRIα-ECD (extracellular domain), as long as the method does not alter a function of the α chain of FcεRI. Such various proteins or peptides may be 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more identical to the amino acid sequence of SEQ ID NO: 1. In addition, the FcεRIα-ECD of SEQ ID NO: 1 may be encoded by a polynucleotide having the sequence of SEQ ID NO: 5.

In addition, as used herein, the term "modified Fc region" means a region in which a part of an Fc portion of an antibody has been modified. Here, the term "Fc region" refers to a protein which contains heavy chain constant region 2 (CH2) and heavy chain constant region 3 (CH3) of an immunoglobulin, and does not contain variable regions of heavy and light chains and light chain constant region 1 (CH1) of an immunoglobulin. In particular, the modified Fc region means a region obtained by substituting some amino acids in the Fc region or by combining different types of Fc regions. Specifically, the modified Fc region may have the amino acid sequence of SEQ ID NO: 2. In addition, the modified Fc region of SEQ ID NO: 2 may be encoded by a polynucleotide having the sequence of SEQ ID NO: 6.

In addition, the "modified Fc region" of the present invention may be in the form of having sugar chains in a native form, increased sugar chains relative to a native form, or decreased sugar chains relative to a native form, or may be in the form of being sugar chain-removed. Immunoglobulin Fc sugar chains may be modified by conventional methods such as chemical methods, enzymatic methods, and genetic engineering methods using microorganisms.

Here, the "modified Fc region" of the present invention may be a region that lacks antibody dependent cellular cytotoxicity (ADCC) and complement dependent cytotoxicity (CDC) functions due to having no binding site for FcγR or C1q. In addition, the modified Fc region and the FcεRIα-ECD may be linked via a hinge of an IgD antibody. The hinge of the IgD antibody is composed of 64 amino acids, and may selectively contain 20 to 60 consecutive amino

acids, 25 to 50 consecutive amino acids, or 30 to 40 amino acids. In an embodiment, the hinge of the IgD antibody may be composed of 30 or 49 amino acids as shown below. In addition, the hinge of the IgD antibody may be a hinge variant obtained by modifying the hinge region, in which the hinge may contain at least one cysteine. Here, the hinge variant may be obtained by modifying some in a hinge sequence of the IgD antibody in order to minimize generation of truncated forms during a protein production process.

In an embodiment, the hinge may contain the following sequence:

Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa1
Xaa2 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr
Lys Thr Pro Glu Cys Pro (SEQ ID NO: 17), where
Xaa1 may be Lys or Gly, and Xaa2 may be Glu, Gly,
or Ser. Specifically, the hinge may have the amino acid
sequence of SEQ ID NO: 3 or SEQ ID NO: 19, thereby
minimizing generation of truncated forms during a
protein production process.

In another embodiment, the hinge may contain the following sequence:

Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr Thr
Ala Pro Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly Glu
Glu Lys Lys Xaa3 Xaa4 Lys Glu Lys Glu Glu Gln Glu
Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro (SEQ ID
NO: 18), where Xaa3 may be Lys or Gly, and Xaa4 may
be Glu, Gly, or Ser. Specifically, the hinge may have the
amino acid sequence of SEQ ID NO: 4, thereby mini-
mizing generation of truncated forms during a protein
production process.

In particular, in the hinge having the sequence of SEQ ID NO: 4, at least one of Thr's may be glycosylated. Specifically, among the amino acids of SEQ ID NO: 18, the 13th, 14th, 18th, or 19th Thr's may be glycosylated. Preferably, all four amino acids may be glycosylated. Here, the glycosylation may be O-glycosylation.

In addition, as described above, the polypeptide dimer provided by the present invention may be in a form in which two monomers are bound to each other and each monomer is obtained by binding between an extracellular domain of an alpha subunit of an IgE Fc receptor and a modified Fc region. The polypeptide dimer may be in a form in which the same two monomers are bound to each other by cysteine located at a hinge site. In addition, the polypeptide dimer may be in a form in which two different monomers are bound to each other. For example, in a case where the two monomers are different from each other, the polypeptide dimer may be in a form in which one monomer contains the extracellular domain of the alpha subunit of the IgE Fc receptor, and the other monomer contains a fragment of the extracellular domain of the alpha subunit of the IgE Fc receptor. Here, an embodiment of the monomer may have the amino acid sequence of SEQ ID NO: 20, SEQ ID NO: 21, or SEQ ID NO: 22.

In addition, the polypeptide dimer provided by the present invention exhibits a binding capacity to IgE which is 10- to 100-fold, 20- to 90-fold, 20- to 70-fold, 30- to 70-fold, or 40- to 70-fold higher than omalizumab, an anti-IgE antibody, and may preferably exhibit a binding capacity to IgE which is 70-fold higher than omalizumab.

In yet another aspect of the invention, there is provided a polynucleotide encoding a monomer that contains an extracellular domain of an alpha subunit of an IgE Fc receptor to which a modified Fc region is bound.

Meanwhile, the polynucleotide may additionally contain a signal sequence or a leader sequence. As used herein, the term "signal sequence" means a nucleic acid encoding a

signal peptide that directs secretion of a target protein. The signal peptide is translated and then cleaved in a host cell. Specifically, the signal sequence of the present invention is a nucleotide encoding an amino acid sequence that initiates protein translocation across the endoplasmic reticulum (ER) membrane. Useful signal sequences in the present invention include antibody light chain signal sequences, for example, antibody 14.18 (Gillies et al., J. Immunol. Meth 1989. 125:191-202), antibody heavy chain signal sequences, for example, the MOPC141 antibody heavy chain signal sequence (Sakano et al., Nature, 1980. 286:676-683), and other signal sequences known in the art (see, for example, Watson et al., Nucleic Acid Research, 1984. 12:5145-5164).

The signal sequences are well known in the art for their characteristics. The signal sequences typically contain 16 to 30 amino acid residues, and may contain greater or fewer amino acid residues. A typical signal peptide consists of three regions which are a basic N-terminal region, a central hydrophobic region, and a more polar C-terminal region. The central hydrophobic region contains 4 to 12 hydrophobic residues that immobilize the signal sequence through the membrane lipid bilayer during translocation of an immature polypeptide.

After initiation, a signal sequence is cleaved in the lumen of ER by cellular enzymes commonly known as signal peptidases. Here, the signal sequence may be a secretory signal sequence for tissue plasminogen activator (tPA), Herpes simplex virus glycoprotein D (HSV gD), or growth hormone. Preferably, secretory signal sequences used in higher eukaryotic cells, including mammals and the like, can be used. In addition, the secretory signal sequence may be substituted with a codon having a high frequency of expression in a host cell, and used.

Meanwhile, the extracellular domain monomer of the alpha subunit of the IgE Fc receptor to which a signal sequence and a modified Fc region are bound, may have the amino acid sequence of SEQ ID NO: 11 or SEQ ID NO: 13. The proteins of SEQ ID NO: 11 and SEQ ID NO: 13 may be encoded by polynucleotides having the sequences of SEQ ID NO: 12 and SEQ ID NO: 14, respectively.

In still yet another aspect of the present invention, there is provided an expression vector loaded with a polynucleotide encoding the monomer. Here, the polynucleotide may have the sequence of SEQ ID NO: 12 or SEQ ID NO: 14.

As used herein, the term "vector" is intended to be introduced into a host cell and to be capable of being recombined with a host cell genome and inserted therein. Alternatively, the vector is an episome and is understood as a nucleic acid unit containing a nucleotide sequence that can be autonomously replicated. The vectors include linear nucleic acids, plasmids, phagemids, cosmids, RNA vectors, viral vectors, and analogs thereof. Examples of the viral vectors include, but are not limited to, retroviruses, adenoviruses, and adeno-associated viruses. In addition, the plasmids may contain a selectable marker such as an antibiotic-resistant gene, and host cells harboring the plasmids may be cultured under selective conditions.

As used herein, the term "genetic expression" or "expression" of a target protein is understood to mean transcription of a DNA sequence, translation of an mRNA transcript, and secretion of a fusion protein product or a fragment thereof. A useful expression vector may be RcCMV (INVITRO-GENTM, Carlsbad) or a variant thereof. The expression vector may contain human cytomegalovirus (CMV) promoter for promoting continuous transcription of a target gene in mammalian cells, and a bovine growth hormone

polyadenylation signal sequence for increasing a stability level of RNA after transcription.

In still yet another aspect of the present invention, there is provided a host cell into which the expression vector is introduced. As used herein, the term "host cell" refers to a prokaryotic or eukaryotic cell into which a recombinant expression vector can be introduced. As used herein, the terms "transduced", "transformed", and "transfected" means introducing a nucleic acid (for example, a vector) into a cell using techniques known in the art.

Preferred host cells that can be used in the present invention include immortal hybridoma cells, NS/O myeloma cells, 293 cells, Chinese hamster ovary cells (CHO cells), HeLa cells, human amniotic fluid-derived cells (CapT cells), or COS cells. Preferably, the host cells may be CHO cells. On the other hand, the host cell may be one in which the vector and a vector loaded with a sialic acid transferase gene are introduced. Here, the sialic acid transferase may be 2,3-sialic acid transferase or 2,6-sialic acid transferase. Here, the 2,6-sialic acid transferase may have the amino acid sequence of SEQ ID NO: 15.

In still yet another aspect of the present invention, there is provided a pharmaceutical composition for treating or preventing an allergic disease, comprising the polypeptide dimer as an active ingredient.

In the present specification, the term "allergic disease" means a pathological symptom caused by an allergic reaction mediated by mast cell activation such as mast cell degranulation. Such allergic diseases include food allergy, atopic dermatitis, asthma, allergic rhinitis, allergic conjunctivitis, allergic dermatitis, allergic contact dermatitis, anaphylaxis, urticaria, pruritus, insect allergy, chronic idiopathic urticaria, drug allergy, and the like. In particular, the allergic diseases may be IgE-mediated.

In the composition for treating or preventing allergic disease of the present invention, an active ingredient may be contained in any amount (effective amount) depending on use, formulation, blending purpose, and the like, as long as the active ingredient can exhibit anti-allergic activity. A typical effective amount of the active ingredient will be determined within a range of 0.001% by weight to 20.0% by weight based on a total weight of the composition. Here, "effective amount" refers to an amount of an active ingredient which is capable of inducing an anti-allergic effect. Such an effective amount can be determined experimentally within the ordinary skill of those skilled in the art.

Here, the pharmaceutical composition may further contain a pharmaceutically acceptable carrier. For the pharmaceutically acceptable carrier, any carrier can be used as long as the carrier is a non-toxic substance suitable for delivery to a patient. Distilled water, alcohol, fat, wax, and an inert solid may be contained as carriers. Pharmaceutically acceptable adjuvants (buffers and dispersants) may also be contained in the pharmaceutical composition.

Specifically, the pharmaceutical composition of the present invention contains, in addition to an active ingredient, a pharmaceutically acceptable carrier, and may be made into an oral or parenteral formulation depending on a route of administration by a conventional method known in the art. Here, the term "pharmaceutically acceptable" means not having more toxicity than a subject to be applied (prescribed) can accommodate without inhibiting activity of the active ingredient.

In a case where the pharmaceutical composition of the present invention is made into an oral formulation, the pharmaceutical composition may be made into formulations such as powders, granules, tablets, pills, sugar coating

tablets, capsules, liquids, gels, syrups, suspensions, and wafers, together with suitable carriers, in accordance with methods known in the art. Here, examples of suitable pharmaceutically acceptable carriers can include sugars such as lactose, glucose, sucrose, dextrose, sorbitol, mannitol, and xylitol, starches such as corn starch, potato starch, and wheat starch, celluloses such as cellulose, methylcellulose, ethylcellulose, sodium carboxymethyl cellulose, and hydroxypropylmethyl cellulose, polyvinyl pyrrolidone, water, methylhydroxybenzoate, propylhydroxybenzoate, magnesium stearate, mineral oil, malt, gelatin, talc, polyol, vegetable oil, and the like. In a case of being made into preparations, the preparations can be carried out, as necessary, by including diluents and/or excipients such as a filler, an extender, a binder, a wetting agent, a disintegrant, and a surfactant.

In a case where the pharmaceutical composition of the present invention is made into a parenteral formulation, the pharmaceutical composition may be made into preparations in the form of injections, transdermal drugs, nasal inhalers, and suppositories, together with suitable carriers, in accordance with methods known in the art. In a case of being prepared into injections, sterilized water, ethanol, polyol such as glycerol and propylene glycol, or a mixture thereof may be used as a suitable carrier. For the carrier, isotonic solutions such as Ringer's solution, phosphate buffered saline (PBS) containing triethanolamine, sterile water for injection, and 5% dextrose, and the like may be preferably used.

Preparation of the pharmaceutical composition is known in the art and specifically, reference can be made to Remington's Pharmaceutical Sciences (19th ed., 1995) and the like. The document is considered part of the present specification.

A preferable daily dosage of the pharmaceutical composition of the present invention is ranged from 0.01 ug/kg to 10 g/kg, and preferably from 0.01 mg/kg to 1 g/kg, depending on the patient's condition, body weight, sex, age, disease severity, or route of administration. Administration may be carried out once or several times a day. Such a dosage should in no way be interpreted as limiting the scope of the present invention.

The subject to which the composition of the present invention can be applied (prescribed) is a mammal and a human, with a human being particularly preferred. The composition for anti-allergy of the present invention may further comprise, in addition to the active ingredient, any compound or natural extract, on which safety has already been verified and which is known to have anti-allergic activity, for the purpose of raising and reinforcing the anti-allergic activity.

In another aspect of the present invention, there is provided a food composition for ameliorating and alleviating an allergic symptom, comprising the polypeptide dimer as an active ingredient.

Here, the polypeptide dimer may be bound to an appropriate delivery unit for efficient delivery into the intestines. The food composition of the present invention may be prepared in any form and may be, for example, prepared in the form of beverages such as tea, juice, carbonated beverage, and ionic beverage, processed dairy products such as milk and yogurt, health functional food preparations such as tablets, capsules, pills, granules, liquids, powders, flakes, pastes, syrups, gels, jellies, and bars, or the like. In addition, the food composition of the present invention may fall within any product category in legal or functional classification as long as the food composition complies with the

enforcement regulations at the time of being manufactured and distributed. For example, the food composition may be a health functional food according to the Health Functional Foods Act, or may fall within confectioneries, beans, teas, beverages, special-purpose foods, or the like according to each food type in the Food Code of Food Sanitation Act (standards and specifications for food, notified by Food and Drug Administration). With regard to other food additives that may be contained in the food composition of the present invention, reference can be made to the Food Code or the Food Additive Code according to the Food Sanitation Act.

In yet another aspect of the present invention, there is provided a method for producing a polypeptide dimer, comprising a step of culturing a host cell into which a polynucleotide encoding a monomer and a sialic acid transferase gene have been introduced; and a step of recovering a polypeptide dimer.

Here, the polynucleotide encoding the monomer may be introduced into the host cell in the form of being loaded on an expression vector. In addition, the sialic acid transferase gene may be introduced into the host cell in the form of being loaded on a vector.

First, a step of introducing, into a host cell, a vector loaded with a polynucleotide encoding a monomer and a vector loaded with a sialic acid transferase gene is carried out. Here, the sialic acid transferase may be 2,3-sialic acid transferase or 2,6-sialic acid transferase.

Next, a step of culturing the transformed cell is carried out.

Finally, a step of recovering a polypeptide dimer is carried out. Here, the polypeptide dimer may be purified from a culture medium or a cell extract. For example, after obtaining supernatant of a culture medium in which the polypeptide dimer is secreted, the supernatant may be concentrated using a commercially available protein concentration filter, for example, an AMICON® or MILLIPORE® PELLICON® ultrafiltration unit. Then, the concentrate may be purified by methods known in the art. For example, the purification may be achieved using a matrix coupled to protein A.

In still yet another aspect of the present invention, there is provided a polypeptide dimer produced by the above-described method for producing a dimer.

Here, the polypeptide dimer has a high sialic acid content, and thus has a very high content of acidic proteins relative to the theoretical pI value.

In still yet another aspect of the present invention, there is provided a pharmaceutical composition for treating or preventing an allergic disease, comprising, as an active ingredient, the polypeptide dimer produced by the above-described method for producing a dimer.

In still yet another aspect of the present invention, there is provided a food composition for ameliorating or alleviating an allergic symptom comprising, as an active ingredient, the polypeptide dimer produced by the above-described method for producing a dimer.

In still yet another aspect of the present invention there is provided a method for treating or preventing an allergic disease, comprising a step of administering to a subject a polypeptide dimer that contains two monomers, each of which contains the extracellular domain (FcεRIα-ECD) of the alpha subunit of the IgE Fc receptor.

The subject may be a mammal, preferably a human. Here, administration may be achieved orally or parenterally. Here, parenteral administration may be performed by methods

such as subcutaneous administration, intravenous administration, mucosal administration, and muscular administration.

Hereinafter, the present invention will be described in more detail with reference to the following examples. However, the following examples are intended to merely illustrate the present invention, and the scope of the present invention is not limited only thereto.

Example 1. Preparation of Polypeptide Containing FcεRIα-ECD and Modified Fc Region

A C-terminal modified polypeptide of the extracellular domain (FcεRIα-ECD) of the alpha subunit of the IgE Fc receptor was prepared according to the method disclosed in U.S. Pat. No. 7,867,491.

First, in order to express a protein (FcεRIαECD-Fc1), a protein (FcεRIαECD-Fc2), and a protein (FcεRIαECD-Fc3), in which the extracellular domain of the α-chain of FcεRI having the amino acid sequence of SEQ ID NO: 1 and the modified immunoglobulin Fc of SEQ ID NO: 2 are linked via a hinge of SEQ ID NO: 19, a hinge of SEQ ID NO: 3, and a hinge of SEQ ID NO: 4, respectively, cassettes obtained by linking the gene encoding each protein were cloned into the pAD15 vectors (Genexin, Inc.) to construct FcεRIαECD-Fc protein expression vectors. Then, each of the expression vectors was transduced into CHO DG44 cells (from Dr. Chasm, Columbia University, USA).

Here, at the time of being transduced into the cell line, an expression vector obtained by cloning an α-2,6-sialic acid transferase gene into the pCI Hygro vector (INVITRO-GEN™) was simultaneously transduced to separately prepare cell lines which are capable of expressing FcεRIαECD-Fc2ST and FcεRIα ECD-Fc3ST proteins to which sialic acid is added.

As a primary screening procedure, HT selection was carried out using 5-hydroxytryptamine (HT)-free 10% dFBS medium (GIBCO™, USA, 30067-334), MEMa medium (GIBCO™, 12561, USA, Cat No. 12561-049), and HT+ medium (GIBCO™, USA, 11067-030). Then, methotrexate (MTX) amplification was performed using HT-selected clones to amplify productivity using the dihydrofolate reductase (DHFR)-system.

After completion of the MTX amplification, subculture was carried out about 1 to 5 times for cell stabilization for the purpose of evaluation of productivity. Thereafter, unit productivity evaluation of the MTX-amplified cells was performed. The results are shown in Table 1 below.

TABLE 1

Version	Media	MTX concentration	Productivity		
			3-day culture		Batch
			ug/mL	ug/10 ⁶ cells	culture (mg/ml)
FcεRIαECD-Fc2	Ex-	500 nM	37.23	20.9	225
FcεRIαECD-Fc2 + a2,6-ST	cell/DHFR	100 nM	45.4	25.1	338.2
FcεRIαECD-Fc3		2 uM	27.0	16.9	180.4
FcεRIαECD-Fc3 + a2,6-ST		1 uM	17.5	10.2	101.7

As shown in Table 1, the FcεRIαECD-Fc3 cell line exhibited productivity of 16.9 ug/10⁶ cells after the methotrexate amplification at 2 uM. On the other hand, the FcεRIαECD-Fc3 cell line (FcεRIαECD-Fc3ST) co-trans-

11

duced with 2,6-sialic acid transferase exhibited productivity of $10.2 \text{ ug}/10^6$ cells after the methotrexate amplification at 1 μM . In addition, the Fc ϵ RI α ECD-Fc2 cell line exhibited productivity of $20.9 \text{ ug}/10^6$ cells under the methotrexate amplification condition at 0.5 μM . In addition, the Fc ϵ RI α ECD-Fc2 cell line (Fc ϵ RI α ECD-Fc2ST) co-transduced with 2,6-sialic acid transferase exhibited productivity of $25.1 \text{ ug}/10^6$ cells after the methotrexate amplification at 0.1 μM . That is, it was identified that the Fc ϵ RI α ECD-Fc2 cell line co-transfected with 2,6-sialic acid transferase, which had been selected under the methotrexate amplification condition at 0.1 μM , exhibits the most excellent productivity.

Example 2. Purification of Fc ϵ RI α ECD Fusion Protein and Identification of Purity Thereof

Among the cell lines selected in Example 1 above, i) Fc ϵ RI α ECD-Fc3, ii) Fc ϵ RI α ECD-Fc3ST, and iii) Fc ϵ RI α ECD-Fc2ST were cultured at a 60 ml scale by a batch culture method. The resulting cultures were purified using a Protein-A affinity column, and then purified proteins were subjected to SDS-PAGE and size-exclusion HPLC (SE-HPLC) to identify purity of the proteins.

As shown in FIGS. 1A-1C, it was identified that all respective proteins purified by the SE-HPLC method have purity of 93% or higher. In addition, as a result of SDS-PAGE analysis, it was identified that proteins having sizes of about 150 kDa and about 75 kDa are detected, respectively, in the non-reducing and reducing conditions (FIG. 1A, Lanes 1 to 6). From this, it was found that the Fc-bound Fc ϵ RI α ECD forms a dimer. In addition, no impurities such as a truncated form were observed in the SDS-PAGE results. In particular, even after the process of thawing/freezing (FIG. 1A, Lanes 7 to 8), it was identified that all proteins have purity of 93% or higher, and has no impurities. From this, it was found that production of truncated protein forms is decreased relative to the Fc ϵ RI α ECD-Fc1 to which the wild type IgD hinge had been applied.

Here, Gel-IEF was performed under the following test conditions to identify a degree of sialic acid content in the proteins following introduction of the sialic acid transferase. From this, it was identified that a content of acidic proteins is increased due to increased sialic acid content.

12

TABLE 2

Test conditions	
Gel	pH3-10 IEF gel 1.0 mm
Sample buffer	IEF sample buffer (2 $\times$)
Loading condition	100 V 1 hr, 200 V 1 hr, 500 V 2 hr

In order to identify reproducibility of purification yield, the Fc ϵ RI α ECD-Fc2ST cell line was batch-cultured in a 1 L flask at a 250 ml scale and purified using a Protein-A affinity column. Subsequently, the culture supernatant and the purified product were subjected to running on a 4% to 15% TGX™ Precast Protein gel (TGX: Tris-Glycine eXtended, Bio-Rad Laboratories, Inc.) for 30 minutes at a condition of Tris-Glycine SDS (TGS) buffer and 200 V, and then subjected to SDS PAGE analysis. As a result, it was identified that not only proteins with very high purity (98% or higher) are purified even by only the first step purification but also proteins are expressed with very high purity even in the culture supernatant. This indicates that process development steps can be simplified in developing the Fc ϵ RI α ECD-Fc protein, which had been expressed in the cell line in question, into a medical product, and as a result, it is highly likely for the development cost of the medical product to be remarkably decreased.

Example 3. Identification of Binding Ability of Fc ϵ RI α ECD Fusion Protein to IgE

A binding ability to IgE was comparatively measured for the four proteins, i) Fc ϵ RI α ECD-Fc2, ii) Fc ϵ RI α ECD-Fc2ST, iii) Fc ϵ RI α ECD-Fc3, and iv) Fc ϵ RI α ECD-Fc3ST which had been purified through the method of Example 2 above, and the commercially available anti-IgE antibody, omalizumab (trade name: XOLAIR®). Specifically, the binding ability to IgE was measured by coating IgE on the channel of the Protein GLC sensor chip (Bio-Rad Laboratories, Inc., Cat #176-5011), and causing omalizumab or each Fc ϵ RI α ECD-Fc protein at various concentrations to flow at a rate of 30 μl per minute.

The experiments were conducted by identifying zero base using 25 mM NaOH as a regeneration buffer, and then repeating the above steps. Thereafter, a binding curve was identified using a protein binding analyzer (PROTEON™ XPR36 protein interaction array system, Bio-Rad Laboratories, Inc., USA). The results are shown in Table 3, and FIGS. 3 and 4.

TABLE 3

Samples items		FcεRIα ECD-Fc		Omalizumab	Remarks
Drug type		Fc fusion protein		Anti-IgE Ab	
Binding affinity	ka (association rate)	Fc2	2.14×10^5	4.05×10^5	1.9-fold weaker than omalizumab
		Fc2ST	2.64×10^5		1.5-fold weaker than omalizumab
		Fc3	1.98×10^5		2.0-fold weaker than omalizumab
		Fc3ST	2.40×10^5		1.7-fold weaker than omalizumab
	kd (dissociation rate)	Fc2	8.29×10^{-5}	6.02×10^{-3}	73-fold better than omalizumab
		Fc2ST	5.69×10^{-5}		106-fold better than omalizumab
		Fc3	1.33×10^{-4}		45-fold better than omalizumab
		Fc3ST	1.49×10^{-4}		40-fold better than omalizumab
	KD (kd/ka)	Fc2	3.88×10^{-10}	1.49×10^{-8}	38-fold better than omalizumab
		Fc2ST	2.16×10^{-10}		69-fold better than omalizumab
		Fc3	6.72×10^{-10}		22-fold better than omalizumab
		Fc3ST	6.21×10^{-10}		24-fold better than omalizumab

13

As shown in Table 3, the association rate (k_a) value of the polypeptide dimer according to an embodiment of the present invention was measured to be 1.5- to 2.0-fold lower than that of omalizumab. That is, it was found that a binding ability thereof to substances other than IgE is 1.5- to 2.0-fold lower than that of omalizumab. In addition, the dissociation rate (k_d) value of the polypeptide dimer according to an embodiment of the present invention was measured to be 40- to 106-fold higher than that of omalizumab. In addition, as shown in FIGS. 3 and 4, it was able to identify that omalizumab loses its binding to IgE in a case where a certain period of time has passed after the binding, whereas once the polypeptide dimer of the Fc ϵ RI α ECD fusion protein of the present invention binds to IgE, the polypeptide dimer is not separated from IgE.

That is, it can be seen that the polypeptide dimer of the present invention is not easily separated from IgE, and has a much better ability to maintain its bound state than omalizumab. As a result, it can be seen that the polypeptide dimer according to an embodiment of the present invention has an equilibrium dissociation constant ($K_D < k_d/k_a$) value which is 22- to 69-fold higher than omalizumab. From this, it was identified that the Fc ϵ RI α ECD fusion protein of the present invention has a remarkably increased binding ability to IgE as compared with omalizumab. In particular, it was identified that the Fc ϵ RI α ECD-Fc2 (Fc ϵ RI α ECD-Fc2ST) to which sialic acid is added exhibits the highest IgE-binding capacity which is 69-fold higher than omalizumab.

Example 4. Identification of Binding Ability of Fc ϵ RI α ECD Fusion Protein to IgG Receptor

A degree of binding of IgE_{TRAP} and omalizumab to IgG receptors was identified using the OCTET® RED384 system (FORTEBIO™, Sartorius AG, CA, USA). Fc γ RI, Fc γ RIIA, Fc γ RIIB, Fc γ RIIIA, and Fc γ RIIIB recombinant proteins (R & D Systems Inc., 5 μ g/ml) were immobilized in 300 mM acetate buffer (pH 5) on the activated AR2G biosensor. As a running buffer, PBS containing 0.1% Tween-20 and 1% bovine serum was used. All experiments were carried out at 30° C. with a sample plate shaker at a rate of 1,000 rpm. The results are shown in FIGS. 5A to 5E, and binding abilities of omalizumab and IgE_{TRAP} to the IgG receptors are quantified and shown in FIG. 6.

Example 5. Identification of Activity of Fc ϵ RI α ECD Fusion Protein Through Beta-Hexosaminidase Assay in Mouse Bone Marrow-Derived Mast Cells

Beta-hexosaminidase assay was performed for in vitro activity analysis of the Fc ϵ RI α ECD fusion protein of the present invention. Specifically, the Fc ϵ RI α ECD-Fc2 protein according to an embodiment of the present invention was mixed, at each concentration, with mouse IgE (1 μ g/mL), and incubated at room temperature (20° C.) for 30 minutes to prepare samples. Mouse bone marrow-derived mast cells in culture for mast cell activation were washed with Hank's balanced salt solution (HBSS) buffer to remove the medium, and the number of cells was measured. Then, an adjustment was made so that 5×10^5 cells were injected into 40 μ L of HBSS buffer.

Then, 50 μ L of the sample solution prepared through the pre-incubation was added to the activated mast cells. Then, the resultant was incubated in a 5% CO₂ incubator at 37° C. for 30 minutes. Subsequently, after the addition of each 10 μ L of DNP (2,4-dinitrophenol, 100 ng/mL), which is a foreign antigen, incubation was performed again at 37° C.

14

for 30 minutes in 5% CO₂, and then 30 μ L of the supernatant was separated. 30 μ L of the separated supernatant and 30 μ L of the substrate (4-nitrophenyl N-acetyl- β -D-glucosaminide, 5.84 mM) were mixed well, and then incubated at 37° C. for 20 minutes in 5% CO₂. Then, 140 μ L of 0.1 M sodium carbonate buffer (pH 10) as a stop solution was added to terminate the reaction. Thereafter, absorbance at 405 nm was measured to identify a secretion amount of β -hexosaminidase secreted by the foreign antigen in the activated mast cells. The results are shown in FIG. 7.

As shown in FIG. 7, the polypeptide dimer of an embodiment of the present invention exhibited a mast cell inhibition ratio of about 49.4% in a case of having half (0.5 μ g/mL) the concentration of mouse IgE, and exhibited a mast cell inhibition ratio of about 99.4% in a case of having the same concentration (1 μ g/mL) of mouse IgE. That is, it can be seen that IgE-induced activity of bone marrow-derived mast cells is greatly suppressed by the Fc ϵ RI α -ECD polypeptide dimer of the present invention.

Example 6. Comparison of Activity of Fc ϵ RI α ECD Fusion Protein and Anti-Human IgE Antibody Using β -Hexosaminidase Assay in Human Fc ϵ RI-Expressing Bone Marrow-Derived Mast Cells

β -Hexosaminidase assay was conducted to identify superiority of the Fc ϵ RI α ECD fusion protein relative to XOLAIR® through in vitro activity analysis. The respective drugs, Fc ϵ RI α ECD-Fc2ST (IgE_{TRAP}) and XOLAIR®, were prepared at each concentration, and then mixed with human IgE (1 μ g/mL). Then, incubation was performed at room temperature for 30 minutes. During pre-incubation of the drug, a human Fc ϵ RI gene was introduced, and mast cells derived from and differentiated from mouse bone marrow, in which the mouse Fc ϵ RI gene had been removed, were prepared. The prepared mast cells were washed with HBSS buffer, and then 5×10^5 cells were injected into 60 μ L of HBSS buffer. 20 μ L of the pre-incubated sample was added to the prepared mast cells, and then incubated in a 5% CO₂ incubator at 37° C. for 30 minutes.

Subsequently, after 20 μ L of anti-human IgE antibody (BioLegend, Inc., Cat No. 325502, 0.5 μ g/mL) was added, and then the resultant was incubated again in 5% CO₂ incubator at 37° C. for 30 minutes. Subsequently, after centrifugation at 1,500 rpm at 4° C., 30 μ L of the supernatant was separated. 30 μ L of the separated supernatant and 30 μ L of the substrate (4-nitrophenyl N-acetyl- μ -glucosaminide, 5.84 mM) were mixed well, and then incubated in a 5% CO₂ incubator at 37° C. for 25 minutes. Then, 140 μ L of 0.1 M sodium carbonate buffer (pH 10) was added to terminate the reaction.

Subsequently, absorbance at 405 nm was measured to compare relative amounts of secreted β -hexosaminidase, and a mass cell-inhibitory effect depending on each drug concentration was identified. The results are shown in FIG. 8. As shown in FIG. 8, IC₅₀ of the Fc ϵ RI α ECD fusion protein was measured to be approximately 11.16 ng/mL, and IC₅₀ of the XOLAIR® protein was measured to be approximately 649.8 ng/mL. Therefore, it was identified that the Fc ϵ RI α ECD fusion protein has a 58-fold higher inhibitory ability on mast cell activity than XOLAIR®.

Example 7. In Vivo Assay of Fc ϵ RI α ECD Fusion Protein (Food Allergy Model)

50 μ g of ovalbumin (OVA) and 1 mg of alum were intraperitoneally administered to Balb/c mice (Orientbio

15

Inc.) two times at a 14-day interval to induce sensitization. Thereafter, 50 mg of OVA was orally administered five times in total on days 28, 30, 32, 34, and 36, to induce food allergy in intestines.

After the OVA was orally administered two times, that is, on day 31, the mice were divided into three groups, each containing 7 mice. The three divided groups were as follows: The first group receiving the FcεRIαECD-Fc2ST fusion protein at a high concentration (200 ug), the second group receiving the FcεRIαECD-Fc2ST fusion protein at a low concentration (20 ug), and the third group receiving nothing. While orally administering the OVA, it was iden-

16

tified whether diarrhea occurs due to food allergy induction. The mice were sacrificed on day 37, and the number of mast cells in the small intestine, the IgE concentration in blood, and the concentration of enzyme (mast cell protease-1 (MCPT-1)) with mast cell degranulation in blood were analyzed for the mice belonging to each group.

As shown in FIG. 9, it was identified that the mice belonging to the group receiving the FcεRIαECD-Fc2ST, which is a polypeptide dimer, at a high concentration exhibits an effect of alleviating food allergy in a concentration-dependent manner, as compared with the mice belonging to the group receiving nothing.

[Sequence List Text]

SEQ ID NO: 1

VPQKPKVSLN PPWNRIFKGE NVT LTCNGNN FFEVSSTKWF HNGSLSEETN SSLNIVNAKF
EDSGEYKQCH QQVNESEPVY LEVFSDWLLL QASAEVVMG QPLFLRCHGW RNWDVYKVIY
YKDGEALKYW YENHNISITN ATVEDSGTYY CTGKVVQLDY ESEPLNITVI KAPREKYWLQ

SEQ ID NO: 2

SHTQPLGVFL FPPKPKDTLM ISRTPEVTCV VVDVSQEDPE VQFNWYVDGV EVHNAKTKPR
EEQFNSTYRV VSVLTVLHQD WLNKEYKCK VSNKGLPSSI EKTISKAKGQ PREPQVYTLF
PSQEEMTKNQ VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFFLYSRLTV
DKSRWQEGNV FSCVMHEAL HNHYTQKSLS LSLGK

SEQ ID NO: 3

RNTGRGGEEK KGSKEKEEQE ERETKTPECP

SEQ ID NO: 4

AQPQAEGLA KATTAPATTR NTGRGGEEKK GSKEKEEQEE RETKTPECP

SEQ ID NO: 5

gtgccccaga agcccaaggt gagcctgaac cctccctgga acagaatctt caagggcgag
aacgtgaccc tgacctgcaa cggcaacaac ttcttcgagg tgagcagcac caagtgggtc
cacaatggca gctgagcga ggagaccaac agctccctga acatcgtgaa cgccaagttc
gaggacagcg gcgagtacaa gtgccagcac cagcaggtga acgagagcga gccctgttac
ctggaggtgt tcacgcactg gctgctgctg caggccagcg ccgaggtggt gatggagggc
cagccccctgt tcctgagatg ccacggctgg agaaactggg acgtgtacaa ggtgatctac
tacaaggatg gcgaggccct gaagtactgg tacgagaacc acaacatctc catcaccaac
gccaccgtgg aggacagcgg cacctactac tgcacaggca aggtgtggca gctggactac
gagagcgagc cctgaacat caccgtgacg aaggctccca gagagaagta ctggctgcag

SEQ ID NO: 6

tgcgtgggtcg tggatgtgag ccaggaagat cccgaagtgc agttcaactg gtacgtggat
ggcgtggaag tgcacaacgc caagaccaag ccagagaag agcagttcaa ctccacctac
agagtgggtga gcgtgctgac cgtgctgcac caggactggc tgaacggcaa ggagtacaag
tgcaaggtgt ccaacaaagg cctgccagc tccatcgaga agaccatcag caaagccaaa
ggccagccca gagaacccca ggtgtacacc ctgcctccca gccaggaaga gatgaccaag
aaccaggtgt cctgacctg cctggtgaaa ggcttctacc ccagcgacat cgccgtggag
tgggaaagca acggccagcc cgagaacaat tacaagacaa cccctcccgt gctggatagc
gatggcagct tctttctgta cagcagactg accgtggaca agagcagatg gcaggaaggg
aacgtgttca gctgcagcgt gatgcacgaa gccctgcaca accactacac ccagaagagc
ctgtccctga gctgggcaa g

-continued

SEQ ID NO: 7

aggaacaccg gcagaggagg cgaggaaaag aaaggaagca aggagaagga ggagcaggag
gaaagagaaa ccaagacccc cgagtgcgcc agccacaccc agcccctggg cgtgttctctg
ttccccccca agcccaagga caccctgatg atcagcagaa cccccgaggt gacc

SEQ ID NO: 8

gcccagcccc aggcgagggg cagcctggct aaggccacca cagctcccgc caccaccagg
aacaccggca gaggaggcga ggaaaagaaa ggaagcaagg agaaggagga gcaggaggaa
agagaaacca agacccccga gtgcccagc cacaccacag cctgggctgt gttcctgttc
cccccaagc ccaaggacac cctgatgatc agcagaacct ccgaggtgac c

SEQ ID NO: 9

MDAMLRGLCC VLLLCGAVFV SPSHA

SEQ ID NO: 10

atggacgcca tgctgagagg cctgtgctgt gtgtgtgtgc tgtgcggcgc cgtgttcgtg
tcccctagcc acgcc

SEQ ID NO: 11

MDAMLRGLCC VLLLCGAVFV SPSHAVPQKP KVS LNPPWNR IFKGENVTLT CNGNNFFEVS
STKWPHNGSL SEETNSSLNI VNAKFEDSGE YKQHQQVNE SEPVYLEVFS DWLLQLASAE
VVMGQPLPL RCHGWRNWDV YKVIYYKDGE ALKYWYENHN ISITNATVED SGTYICTGKV
WQLDYESEPL NITVIKAPRE KYWLQRNTGR GGEKKGSKE KEEQEERETK TPECPSTQP
LGVFLFPPKP KDTLMISRTPEVTCVVVDVS QEDPEVQFNW YVDGVEVHNA KTKPREEQFN
STYRVVSVLT VLHQDWLNGK EYKCKVSNKG LPSSIEKTIS KAKGQPREPQ VYTLPPSQEE
MTKNQVSLTC LVKGFYP SDIAVEWESNGQP ENNYKTTTPV LDSDGSFFLY SRLTVDKSRW
QEGNVFSCSV MHEALHNHYT QKSLSLSLGK

SEQ ID NO: 12

atggacgcca tgctgagagg cctgtgctgt gtgtgtgtgc tgtgcggcgc cgtgttcgtg
tcccctagcc acgcggtgcc ccagaagccc aaggtagacc tgaacctcc ctggaacaga
atcttcaagg gcgagaacgt gacctgacc tgcaacggca acaattctt cgaggtgagc
agcaccaagt ggttcacaa tggcagcctg agcgaggaga ccaacagctc cctgaacatc
gtgaacgcca agttcgagga cagcggcgag tacaagtgcc agcaccagca ggtgaacgag
agcgagcccc tgtacctgga ggtgttcagc gactggctgc tgctgcaggc cagcgccgag
gtggtgatgg agggccagcc cctgttctct agatgccacg gctggagaaa ctgggacgtg
tacaaggtag tctactacaa ggatggcgag gccctgaagt actggtacga gaaccacaac
atctccatca ccaacgccac cgtggaggac agcggcacct actactgcac aggaaggtg
tggcagctgg actacgagag cgagccccctg aacatcaccc tgatcaaggc tcccagagag
aagtactggc tgcagaggaa caccggcaga ggaggcgagg aaaagaaagg aagcaaggag
aaggaggagc agggaggaaag agaaaccaag acccccagat gcccagcca caccagccc
ctgggctgtg tctgttctcc ccccaagccc aaggacaccc tgatgatcag cagaaccccc
gaggtgacct gcgtggtcgt ggatgtgagc caggaagatc ccgaagtga gttcaactgg
tacgtggatg gcgtggaagt gcacaacgcc aagaccaagc ccagagaaga gcagttaac
tccacctaca gagtggtag cgtgtgtgacc gtgtgtgacc aggactggct gaacggcaag
gagtacaagt gcaagggtgc caacaaaggc ctgcccagct ccatcgagaa gaccatcagc
aaagccaaag gccagcccag agaaccacag gtgtacaccc tgctccccag ccaggaagag
atgaccaaga accaggtgtc cctgacctgc ctggtgaaag gcttctaccc cagcgacatc

- continued

gccgtggagt gggaaagcaa cggccagccc gagaacaatt acaagacaac ccctcccgtg
 ctggatagcg atggcagctt ctttctgtac agcagactga ccgtggacaa gagcagatgg
 caggaaggca acgtgttcag ctgcagcgtg atgcacgaag ccctgcacaa ccactacacc
 cagaagagcc tgtccctgag cctgggcaag

SEQ ID NO: 13

MDAMLRGLCC VLLLCGAVFV SPSHAVPQKP KVS LNPPWNR IFKGENVTLT CNGNNFFEVS
 STKWFHNGSL SEETNSSLNI VNAKFEDSGE YKCQHQQVNE SEPVYLEVFS DWLLQLASAE
 VVMEGQPLFL RCHGWRNWDV YKVIYYKDGE ALKYWYENHN ISITNATVED SGTYICTGKV
 WQLDYESEPL NITVIKAPRE KYWLQAQPQA EGSLAKATTA PATTRNTGRG GEEKKGSKEK
 EEQEERETKT PECPSHTQPL GVFLFPPKPK DTLMISRTPE VTCVVVDVSQ EDPEVQFNWY
 VDGVEVHNKAK TKPREEQFNS TYRVVSVLTV LHQDWLNGKE YKCKVSNKGL PSSIEKTISK
 AKGQPREPQV YTLPPSQEEM TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL
 DSDGSFFLYS RLTVDKSRWQ EGNVFSCSVN HEALHNHYTQ KSLSLSLGK

SEQ ID NO: 14

atggacgcca tgcctgagagg cctgtgctgt gtgctgctgc tgtgcggcgc cgtgttcgtg
 tcccctagcc acgccgtgcc ccagaagccc aaggtagacc tgaaccctcc ctggaacaga
 atcttcaagg gcgagaacgt gaccctgacc tgcaacggca acaacttctt cgaggtaggc
 agcaccaagt ggttcacaa tggcagcctg agcgaggaga ccaacagctc cctgaacatc
 gtgaacgcca agttcgagga cagcggcgag tacaagtgcc agcaccagca ggtgaacgag
 agcgagcccg tgtacctgga ggtgttcagc gactggctgc tgcctgaggc cagcgccgag
 gtggtgatgg agggccagcc cctgttctct agatgccacg gctggagaaa ctgggacgtg
 tacaaggtag tctactacaa ggatggcgag gccctgaagt actggtacga gaaccacaac
 atctccatca ccaacgccac cgtggaggac agcggcacct actactgcac aggcaagggtg
 tggcagctgg actacagag cgagcccctg aacatcaccc tgatcaaggc tcccagagag
 aagtactggc tgcaggccca gcccagggc gagggcagcc tggctaaggc caccacagct
 cccgccacca ccaggaaacac cggcagagga ggcgagggaaa agaaaggaaag caaggagaag
 gaggagcagg aggaagagaga aaccaagacc cccgagtgcc ccagccacac ccagcccctg
 ggcgtgttcc tgttcccccc caagcccaag gacaccctga tgatcagcag aacccccgag
 gtgacctgcg tggctgtgga tgtgagccag gaagatcccc aagtgcagtt caactggtac
 gtggatggcg tggaagtgca caacgccaag accaagccca gagaagagca gttcaactcc
 acctacagag tggtagagcgt gctgaccgtg ctgcaccagg actggctgaa cggcaaggag
 tacaagtgca aggtgtccaa caaaggcctg cccagctcca tcgagaagac catcagcaaa
 gccaaaggcc agcccagaga accccaggtg tacaccctgc ctcccagcca ggaagagatg
 accaagaacc aggtgtccct gacctgcctg gtgaaaggct tctaccccag cgacatcgcc
 gtggagtggg aaagcaacgg ccagcccag aacaattaca agacaacccc tcccgtgctg
 gatagcagat gcagcttctt tctgtacagc agactgaccg tggacaagag cagatggcag
 gaaggaacg tgttcagctg cagcgtgatg cacgaagccc tgcacaacca ctacaccag
 aagagcctgt ccctgagcct gggcaag

SEQ ID NO: 15

MIHTNLKKKF SCCVLVFLLF AVICVWKEKK KGSYYDSFKL QTKFQVLKS LGKLMGSDS
 QSVSSSTQD PHRGRQTLGS LRGLAKAKPE ASFQVWNKDS SSKNLIPRLQ KIWKNYLSMN
 KYKVSYKGGP GIKFSAEAL RCHLRDHVNV SMVEVTDFFP NTSEWEGYLP KESIRTKAGP

-continued

WGRCVVSSA GSKSSQLGR EIDHDHDAVLR FNGAPTANFQ QDVGKTITIR LMNSQLVTTE
 KRFLKDSLYN EGILIVWDPS VYHSDIPKWY QNPDYNFFNN YKTYRKLHPN QPFYILKPQM
 PWELWDILQE ISPEEIQPNP PSSGMLGIII MMTLCDQVDI YEFLPSKRKT DVCYYYQKFF
 DSACTMGAYH PLYEKNLVK HLNQGTDEDI YLLGKATLPG FRTIHC

SEQ ID NO: 16

atgatccaca ccaacctgaa gaagaagttc agctgctgcg tgctggtgtt cctgctgttc
 gccgtgatct gcgtgtggaa ggagaagaag aaaggcagct actacgacag ctccaagctg
 cagaccaagg agttccaggt gctgaagagc ctgggcaagc tggccatggg cagcgacagc
 cagagcgtgt ccagctcctc caccaggat cccacagag gcagacagac cctgggcagc
 ctgagaggcc tggccaaggc caagcccag gccagcttcc aggtgtggaa caaggacagc
 agcagcaaga acctgatccc cagactgcag aagatctgga agaactacct gagcatgaac
 aagtacaagg tgagctacaa aggaccgga cccggcatca agttcagcgc cgaggccctg
 aggtgccacc tgagagacca cgtgaacgtg agcatggtgg aagtgaccga ctcccccttc
 aacaccagcg agtgggaagg ctacctgccc aaggagagca tcaggaccaa ggctggcccc
 tggggcagat gcgccgtggt gagcagcgtt ggcagcctga agagctccca gctgggcaga
 gagatcgacg accacgatgc cgtgctgagg ttcaatggcg ctcccaccgc caacttcag
 caggacgtgg gcaccaagac cacaatccgg ctgatgaaca gccagctggt gacaaccgag
 aagcggttcc tgaaggacag cctgtacaac gagggcatcc tgatcgtgtg ggatcccagc
 gtgtaccaca gcgacatccc caagtggtag cagaatcccg actacaactt ctccaacaac
 tacaagacct atagaaagct gcaccccaac cagcccttct acatcctgaa gcccagatg
 ccctgggagc tgtgggacat cctgcaggag atcagccctg aagagatcca gcccaccct
 ccctccagcg gcctgctggg cattatcctc atgatgacct tgtgcgacca ggtggacatc
 tacgagttcc tgcccagcaa gagaaagacc gacgtgtgct actactatca gaagtcttc
 gacagcgctt gcaccatggg cgccctaccac cccctgctgt acgagaagaa cctggtgaag
 cacctgaacc agggcaccga cgaggacatc tacctgctgg gcaaagccac cctgcccggc
 ttcagaacca tccactgc

SEQ ID NO: 17

RNTGRGEEK KXXKEKEEQE ERETKTPECP

SEQ ID NO: 18

AQPQAEGLA KATTAPATTR NTGRGEEKK XXKEKEEQEE RETKTPECP

SEQ ID NO: 19

RNTGRGEEK KKEKEKEEQE ERETKTPECP

SEQ ID NO: 20

VPQKPKVSLN PPWNRIKGE NVTILTCNGNN FFEVSSTKWF HNGSLSEETN SSLNIVNAKF
 EDSGEYKCQH QQVNESEPVY LEVFSDWLLL QASAEVVMG QPLFLRCHGW RNWDVYKVIY
 YKDGEALKYW YENHNISITN ATVEDSGTYY CTGKVVQLDY ESEPLNITVI KAPREKYWLQ
 RNTGRGEEK KKEKEKEEQE ERETKTPECP SHTQPLGVFL FPPKPKDTLM ISRTPEVTCV
 VVDVSQEDPE VQFNWYVDGV EVHNAKTKPR EEQFNSTYRV VSVLTVLHQD WLNKEYKCK
 VSNKGLPSSI EKTISKAKGQ PREPQVYTL PSEQEMTKNQ VSLTCLVKGF YPSDIAVEWE
 SNGQPENNYK TTPPVLDSDG SFFLYSRLTV DKSRWQEGNV FSCSVMEAL HNHYTQKSL
 LSLGK

SEQ ID NO: 21

VPQKPKVSLN PPWNRIKGE NVTILTCNGNN FFEVSSTKWF HNGSLSEETN SSLNIVNAKF
 EDSGEYKCQH QQVNESEPVY LEVFSDWLLL QASAEVVMG QPLFLRCHGW RNWDVYKVIY

-continued

YKDGEALKYW YENHNISITN ATVEDSGTYY CTGKVVQLDY ESEPLNITVI KAPREKYWLQ
 RNTGRGGEK KGSKEKEEQE ERETKTPECP SHTQPLGVFL FPPKPKDTLM ISRTPEVTCV
 VVDVSQEDPE VQFNWYVDGV EVHNAKTKPR EEQFNSTYRV VSVLTVLHQD WLNGKEYKCK
 VSNKGLPSSI EKTISKAKGQ PREPQVYTLF PSQEEMTKNQ VSLTCLVKGF YPSDIAVEWE
 SNGQPENNYK TTPPVLDSDG SFFLYSRLTV DKSRWQEGNV FSCSVMEAL HNHYTQKSL
 LSLGK

SEQ ID NO: 22

VPQKPKVSLN PPWNRIKGE NVTLCNGNN FFEVSSTKWF HNGSLSEETN SSLNIVNAKF
 EDSGEYKQCH QQVNESEPVY LEVFSDWLLL QASAEVVMG QPLFLRCHGW RNWDVYKVIY
 YKDGEALKYW YENHNISITN ATVEDSGTYY CTGKVVQLDY ESEPLNITVI KAPREKYWLQ
 AQPQAEGLA KATTAPATTR NTGRGGEKK GSKEKEEQEE RETKTPECP HTQPLGVFLF
 PPKPKDTLMI SRTPEVTCV VVDVSQEDPEV QFNWYVDGVE VHNKTKPRE EQFNSTYRVV
 SVLTVLHQDW LNGKEYKCKV SNKGLPSSIE KTISKAKGQP REPQVYTLPP SQEEMTKNQV
 SLTCLVKGFY PSDIAVEWES NGQPENNYKT TTPPVLDSDG SFFLYSRLTV DKSRWQEGNV
 FSCSVMEALH NHYTQKSLSL SLGK

 SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 22

<210> SEQ ID NO 1

<211> LENGTH: 180

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: FCeR11 ECD

<400> SEQUENCE: 1

Val Pro Gln Lys Pro Lys Val Ser Leu Asn Pro Pro Trp Asn Arg Ile
 1 5 10 15
 Phe Lys Gly Glu Asn Val Thr Leu Thr Cys Asn Gly Asn Asn Phe Phe
 20 25 30
 Glu Val Ser Ser Thr Lys Trp Phe His Asn Gly Ser Leu Ser Glu Glu
 35 40 45
 Thr Asn Ser Ser Leu Asn Ile Val Asn Ala Lys Phe Glu Asp Ser Gly
 50 55 60
 Glu Tyr Lys Cys Gln His Gln Gln Val Asn Glu Ser Glu Pro Val Tyr
 65 70 75 80
 Leu Glu Val Phe Ser Asp Trp Leu Leu Leu Gln Ala Ser Ala Glu Val
 85 90 95
 Val Met Glu Gly Gln Pro Leu Phe Leu Arg Cys His Gly Trp Arg Asn
 100 105 110
 Trp Asp Val Tyr Lys Val Ile Tyr Tyr Lys Asp Gly Glu Ala Leu Lys
 115 120 125
 Tyr Trp Tyr Glu Asn His Asn Ile Ser Ile Thr Asn Ala Thr Val Glu
 130 135 140
 Asp Ser Gly Thr Tyr Tyr Cys Thr Gly Lys Val Trp Gln Leu Asp Tyr
 145 150 155 160
 Glu Ser Glu Pro Leu Asn Ile Thr Val Ile Lys Ala Pro Arg Glu Lys
 165 170 175
 Tyr Trp Leu Gln

-continued

180

<210> SEQ ID NO 2
 <211> LENGTH: 215
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Modified Fc

<400> SEQUENCE: 2

Ser His Thr Gln Pro Leu Gly Val Phe Leu Phe Pro Pro Lys Pro Lys
 1 5 10 15
 Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val
 20 25 30
 Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp
 35 40 45
 Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe
 50 55 60
 Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp
 65 70 75 80
 Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu
 85 90 95
 Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg
 100 105 110
 Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu Met Thr Lys
 115 120 125
 Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp
 130 135 140
 Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys
 145 150 155 160
 Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser
 165 170 175
 Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn Val Phe Ser
 180 185 190
 Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
 195 200 205
 Leu Ser Leu Ser Leu Gly Lys
 210 215

<210> SEQ ID NO 3
 <211> LENGTH: 30
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgD hinge variant

<400> SEQUENCE: 3

Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Gly Ser Lys Glu Lys
 1 5 10 15
 Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro
 20 25 30

<210> SEQ ID NO 4
 <211> LENGTH: 49
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgD hinge variant

<400> SEQUENCE: 4

-continued

Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr Thr Ala Pro
1 5 10 15

Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Gly Ser
20 25 30

Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys
35 40 45

Pro

<210> SEQ ID NO 5
<211> LENGTH: 540
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: nucleotides sequence of FCeRI1 ECD

<400> SEQUENCE: 5

```
gtgccccaga agcccaaggt gagcctgaac cctccctgga acagaatctt caagggcgag      60
aacgtgacct tgacctgcaa cggaacaac ttcttcgagg tgagcagcac caagtgggtc      120
cacaatggca gcctgagcga ggagaccaac agctccctga acatcgtgaa cgccaagttc      180
gaggacagcg gcgagtacaa gtgccagcac cagcaggtga acgagagcga gcccggtgac      240
ctggagggtg tcagcgactg gctgctgctg caggccagcg ccgagggtgt gatggagggc      300
cagccccctg tcctgagatg ccacggctgg agaaactggg acgtgtacaa ggtgatctac      360
tacaaggatg gcgaggccct gaagtactgg tacgagaacc acaacatctc catcaccaac      420
gccaccgtgg aggacagcgg cacctactac tgcacaggca aggtgtggca gctggactac      480
gagagcgagc ccctgaacat caccgtgatc aaggctccca gagagaagta ctggctgcag      540
```

<210> SEQ ID NO 6
<211> LENGTH: 561
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: nucleotides sequence of modified Fc

<400> SEQUENCE: 6

```
tgctggtgtg tggatgtgag ccaggaagat cccgaagtgc agttcaactg gtacgtggat      60
ggcgtggaag tgcacaacgc caagaccaag cccagagaag agcagttcaa ctccacctac      120
agagtgggtg gcgtgctgac cgtgctgcac caggactggc tgaacggcaa ggagtacaag      180
tgcaagggtg ccaacaaagg cctgcccagc tccatcgaga agaccatcag caaagccaaa      240
ggccagccca gagaacccca ggtgtacacc ctgcctccca gccaggaaga gatgaccaag      300
aaccaggtgt ccctgacctg cctggtgaaa ggcttctacc ccagcgacat cgccgtggag      360
tggaagagca acgcccagcc cgagaacaat tacaagacaa cccctcccgt gctggatagc      420
gatggcagct tctttctgta cagcagactg accgtggaca agagcagatg gcaggaaggc      480
aacgtgttca gctgcagcgt gatgcacgaa gccctgcaca accactacac ccagaagagc      540
ctgtccctga gcctgggcaa g                                     561
```

<210> SEQ ID NO 7
<211> LENGTH: 174
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: nucleotides sequence of IgD hinge variant

<400> SEQUENCE: 7

-continued

```

aggaacaccg gcagaggagg cgaggaaaag aaaggaagca aggagaagga ggagcaggag      60
gaaagagaaa ccaagacccc cgagtgcccc agccacaccc agccccctggg cgtgttctctg    120
ttccccccca agcccaagga caccctgatg atcagcagaa cccccgaggt gacc              174

```

```

<210> SEQ ID NO 8
<211> LENGTH: 231
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: nucleotides sequence of IgD hinge variant

```

```

<400> SEQUENCE: 8
gccacgcccc aggccgaggg cagcctggct aaggccacca cagctccgc caccaccagg      60
aacaccggca gaggaggcga ggaaaagaaa ggaagcaagg agaaggagga gcaggaggaa    120
agagaaacca agacccccga gtgccccagc cacaccagc ccctgggcgt gttcctgttc    180
ccccccaagc ccaaggacac cctgatgatc agcagaacct ccgaggtgac c              231

```

```

<210> SEQ ID NO 9
<211> LENGTH: 25
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: signal peptide

```

```

<400> SEQUENCE: 9

```

```

Met Asp Ala Met Leu Arg Gly Leu Cys Cys Val Leu Leu Leu Cys Gly
1      5              10              15
Ala Val Phe Val Ser Pro Ser His Ala
      20              25

```

```

<210> SEQ ID NO 10
<211> LENGTH: 75
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: nucleotides sequence of signal peptide

```

```

<400> SEQUENCE: 10
atggacgcca tgctgagagg cctgtgctgt gtgctgctgc tgtgcgcgcc cgtgttcgtg    60
tccctagcc acgcc              75

```

```

<210> SEQ ID NO 11
<211> LENGTH: 450
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: FceRIa ECD-hinge-Fc2

```

```

<400> SEQUENCE: 11

```

```

Met Asp Ala Met Leu Arg Gly Leu Cys Cys Val Leu Leu Leu Cys Gly
1      5              10              15
Ala Val Phe Val Ser Pro Ser His Ala Val Pro Gln Lys Pro Lys Val
      20              25              30
Ser Leu Asn Pro Pro Trp Asn Arg Ile Phe Lys Gly Glu Asn Val Thr
      35              40              45
Leu Thr Cys Asn Gly Asn Asn Phe Phe Glu Val Ser Ser Thr Lys Trp
      50              55              60
Phe His Asn Gly Ser Leu Ser Glu Glu Thr Asn Ser Ser Leu Asn Ile
      65              70              75              80

```

Val	Asn	Ala	Lys	Phe	Glu	Asp	Ser	Gly	Glu	Tyr	Lys	Cys	Gln	His	Gln
85															
Gln	Val	Asn	Glu	Ser	Glu	Pro	Val	Tyr	Leu	Glu	Val	Phe	Ser	Asp	Trp
100															
Leu	Leu	Leu	Gln	Ala	Ser	Ala	Glu	Val	Val	Met	Glu	Gly	Gln	Pro	Leu
115															
Phe	Leu	Arg	Cys	His	Gly	Trp	Arg	Asn	Trp	Asp	Val	Tyr	Lys	Val	Ile
130															
Tyr	Tyr	Lys	Asp	Gly	Glu	Ala	Leu	Lys	Tyr	Trp	Tyr	Glu	Asn	His	Asn
145															
Ile	Ser	Ile	Thr	Asn	Ala	Thr	Val	Glu	Asp	Ser	Gly	Thr	Tyr	Tyr	Cys
165															
Thr	Gly	Lys	Val	Trp	Gln	Leu	Asp	Tyr	Glu	Ser	Glu	Pro	Leu	Asn	Ile
180															
Thr	Val	Ile	Lys	Ala	Pro	Arg	Glu	Lys	Tyr	Trp	Leu	Gln	Arg	Asn	Thr
195															
Gly	Arg	Gly	Gly	Glu	Glu	Lys	Lys	Gly	Ser	Lys	Glu	Lys	Glu	Glu	Gln
210															
Glu	Glu	Arg	Glu	Thr	Lys	Thr	Pro	Glu	Cys	Pro	Ser	His	Thr	Gln	Pro
225															
Leu	Gly	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile
245															
Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Gln	Glu
260															
Asp	Pro	Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His
275															
Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Tyr	Arg
290															
Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys
305															
Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ser	Ser	Ile	Glu
325															
Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr
340															
Thr	Leu	Pro	Pro	Ser	Gln	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu
355															
Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp
370															
Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val
385															
Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Arg	Leu	Thr	Val	Asp
405															
Lys	Ser	Arg	Trp	Gln	Glu	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His
420															
Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Leu
435															
Gly	Lys														
450															

```
<210> SEQ ID NO 12
<211> LENGTH: 1350
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
```

-continued

```

<223> OTHER INFORMATION: nucleotides sequence of FcεRIα ECD-hinge-Fc2

<400> SEQUENCE: 12

atggacgcc tgcagagagg cctgtgctgt gtgctgctgc tgtgcggcgc cgtgttcgtg      60
tcccctagcc acgccgtgcc ccagaagccc aaggtgagcc tgaacctccc ctggaacaga      120
atcttcaagg gcgagaacgt gaccctgacc tgcaacggca acaacttctt cgaggtgagc      180
agcaccaagt ggttcacaaa tggcagcctg agcgaggaga ccaacagctc cctgaacatc      240
gtgaacgcc agttcgagga cagcggcgag tacaagtgcc agcaccagca ggtgaacgag      300
agcgagcccc tgtacctgga ggtgttcagc gactggctgc tgcctcaggc cagcgccgag      360
gtggtgatgg agggccagcc cctgttctct agatgccacg gctggagaaa ctgggacgtg      420
tacaaggtag tctactacaa ggatggcgag gccctgaagt actggtacga gaaccacaac      480
atctccatca ccaacgccac cgtggaggac agcggcacct actactgcac aggcaaggtg      540
tggcagctgg actacgagag cgagccctct aacatcaccc tgatcaaggc tcccagagag      600
aagtactggc tgcagaggaa caccggcaga ggaggcgagg aaaagaaagg aagcaaggag      660
aaggaggagc aggaggaaag agaaccaag acccccagct gcccagcca caccagccc      720
ctgggcgtgt tcctgttccc ccccaagccc aaggaccccc tgatgatcag cagaaccccc      780
gaggtgacct gcgtggtcgt ggatgtgagc caggaagatc ccgaagtgca gttcaactgg      840
tacgtggatg gcgtggaagt gcacaacgcc aagaccaagc ccagagaaga gcagttcaac      900
tccacctaca gagtggtagc cgtgtgacc gtgctgcacc aggactggct gaacggcaag      960
gagtacaagt gcaagggtgc caacaaaggc ctgcccagct ccatcgagaa gaccatcagc     1020
aaagccaaag gccagcccag agaaccacag gtgtaccccc tgctcccag ccaggaagag     1080
atgaccaaga accaggtgtc cctgacctgc ctgggtgaaag gcttctaccc cagcgacatc     1140
gccgtggagt gggaaagcaa cgccagccc gagaacaatt acaagacaac ccctcccgctg     1200
ctggatagcg atggcagctt ctttctgtac agcagactga ccgtggacaa gagcagatgg     1260
caggaaggca acgtgttcag ctgcagcgtg atgcacgaag ccctgcacaa ccactacacc     1320
cagaagagcc tgtccctgag cctgggcaag                                     1350

```

```

<210> SEQ ID NO 13
<211> LENGTH: 469
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: FcεRIα ECD-hinge-Fc3

```

```

<400> SEQUENCE: 13

Met Asp Ala Met Leu Arg Gly Leu Cys Cys Val Leu Leu Leu Cys Gly
1           5           10          15

Ala Val Phe Val Ser Pro Ser His Ala Val Pro Gln Lys Pro Lys Val
20          25          30

Ser Leu Asn Pro Pro Trp Asn Arg Ile Phe Lys Gly Glu Asn Val Thr
35          40          45

Leu Thr Cys Asn Gly Asn Asn Phe Phe Glu Val Ser Ser Thr Lys Trp
50          55          60

Phe His Asn Gly Ser Leu Ser Glu Glu Thr Asn Ser Ser Leu Asn Ile
65          70          75          80

Val Asn Ala Lys Phe Glu Asp Ser Gly Tyr Lys Cys Gln His Gln
85          90          95

Gln Val Asn Glu Ser Glu Pro Val Tyr Leu Glu Val Phe Ser Asp Trp

```

-continued

100					105					110					
Leu	Leu	Leu	Gln	Ala	Ser	Ala	Glu	Val	Val	Met	Glu	Gly	Gln	Pro	Leu
		115					120					125			
Phe	Leu	Arg	Cys	His	Gly	Trp	Arg	Asn	Trp	Asp	Val	Tyr	Lys	Val	Ile
	130					135					140				
Tyr	Tyr	Lys	Asp	Gly	Glu	Ala	Leu	Lys	Tyr	Trp	Tyr	Glu	Asn	His	Asn
145					150					155					160
Ile	Ser	Ile	Thr	Asn	Ala	Thr	Val	Glu	Asp	Ser	Gly	Thr	Tyr	Tyr	Cys
				165					170					175	
Thr	Gly	Lys	Val	Trp	Gln	Leu	Asp	Tyr	Glu	Ser	Glu	Pro	Leu	Asn	Ile
			180					185					190		
Thr	Val	Ile	Lys	Ala	Pro	Arg	Glu	Lys	Tyr	Trp	Leu	Gln	Ala	Gln	Pro
		195					200					205			
Gln	Ala	Glu	Gly	Ser	Leu	Ala	Lys	Ala	Thr	Thr	Ala	Pro	Ala	Thr	Thr
	210					215					220				
Arg	Asn	Thr	Gly	Arg	Gly	Gly	Glu	Glu	Lys	Lys	Gly	Ser	Lys	Glu	Lys
225					230					235					240
Glu	Glu	Gln	Glu	Glu	Arg	Glu	Thr	Lys	Thr	Pro	Glu	Cys	Pro	Ser	His
				245					250					255	
Thr	Gln	Pro	Leu	Gly	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr
			260					265						270	
Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val
		275					280					285			
Ser	Gln	Glu	Asp	Pro	Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val
	290					295					300				
Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser
305					310					315					320
Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu
				325					330					335	
Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ser
			340					345					350		
Ser	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro
	355						360					365			
Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Gln	Glu	Glu	Met	Thr	Lys	Asn	Gln
	370					375					380				
Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala
385					390					395					400
Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr
				405					410					415	
Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Arg	Leu
			420					425					430		
Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Glu	Gly	Asn	Val	Phe	Ser	Cys	Ser
		435					440					445			
Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser
						455					460				
Leu	Ser	Leu	Gly	Lys											
465															

<210> SEQ ID NO 14

<211> LENGTH: 1407

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: nucleotides sequence of FcεRIα ECD-hinge-Fc3

-continued

<400> SEQUENCE: 14

```

atggacgcc  tgcctgagagg  cctgtgctgt  gtgctgctgc  tgtgcggcgc  cgtgttcgtg      60
ccccctagcc  acgcogtgcc  ccagaagccc  aaggtgagcc  tgaaccctcc  ctggaacaga      120
atcttcaagg  gcgagaacgt  gacctgacc  tgcaacggca  acaacttctt  cgaggtgagc      180
agcaccaagt  ggttcacaa  tggcagcctg  agcgaggaga  ccaacagctc  cctgaacatc      240
gtgaacgcc  agttcgagga  cagcggcgag  tacaagtgcc  agcaccagca  ggtgaacgag      300
agcgagcccg  tgtacctgga  ggtgttcagc  gactggctgc  tgctgcaggc  cagcgccgag      360
gtggtgatgg  agggccagcc  cctgttctct  agatgccacg  gctggagaaa  ctgggacgtg      420
tacaagtgta  tctactacaa  ggatggcgag  gccctgaagt  actggtacga  gaaccacaac      480
atctccatca  ccaacgccac  cgtggaggac  agcggcacct  actactgcac  aggcaaggtg      540
tggcagctgg  actacgagag  cgagccctg  aacatcaccc  tgatcaaggc  tcccagagag      600
aagtactggc  tgcaggccca  gccccaggcc  gagggcagcc  tggctaaggc  caccacagct      660
ccccccacca  ccaggaacac  cggcagagga  ggcgagggaa  agaaaggaag  caaggagaag      720
gaggagcagg  aggaaagaga  aaccaagacc  cccgagtgcc  ccagccacac  ccagccctg      780
ggcgtgttcc  tgttcccccc  caagcccaag  gacacctga  tgatcagcag  aacccccgag      840
gtgacctgcg  tggctgtgga  tgtgagccag  gaagatcccg  aagtgcagtt  caactggtac      900
gtggatggcg  tggaagtgca  caacgccaag  accaagccca  gagaagagca  gttcaactcc      960
acctacagag  tggtagcgct  gctgaccgtg  ctgcaccagg  actggtgtaa  cggcaaggag     1020
tacaagtgca  aggtgtccaa  caaaggcctg  cccagctcca  tcgagaagac  catcagcaaa     1080
gccaagggcc  agcccagaga  accccagggt  tacacctgct  ctcccagcca  ggaagagatg     1140
accaagaacc  aggtgtccct  gacctgctg  gtgaaaggct  tctaccccag  cgacatcgcc     1200
gtggagtggg  aaagcaacgg  ccagcccgag  aacaattaca  agacaacccc  tcccgtgctg     1260
gatagcgatg  gcagcttctt  tctgtacagc  agactgaccg  tggacaagag  cagatggcag     1320
gaaggcaacg  tgttcagctg  cagcgtgatg  cacgaagccc  tgcacaacca  ctacaccag      1380
aagagcctgt  ccctgagcct  gggcaag                                1407

```

<210> SEQ ID NO 15

<211> LENGTH: 406

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: human a-2,6 sialic acid transferase

<400> SEQUENCE: 15

```

Met Ile His Thr Asn Leu Lys Lys Lys Phe Ser Cys Cys Val Leu Val
1           5           10          15

Phe Leu Leu Phe Ala Val Ile Cys Val Trp Lys Glu Lys Lys Lys Gly
20          25          30

Ser Tyr Tyr Asp Ser Phe Lys Leu Gln Thr Lys Glu Phe Gln Val Leu
35          40          45

Lys Ser Leu Gly Lys Leu Ala Met Gly Ser Asp Ser Gln Ser Val Ser
50          55          60

Ser Ser Ser Thr Gln Asp Pro His Arg Gly Arg Gln Thr Leu Gly Ser
65          70          75          80

Leu Arg Gly Leu Ala Lys Ala Lys Pro Glu Ala Ser Phe Gln Val Trp
85          90          95

Asn Lys Asp Ser Ser Ser Lys Asn Leu Ile Pro Arg Leu Gln Lys Ile

```

-continued

100							105					110				
Trp	Lys	Asn	Tyr	Leu	Ser	Met	Asn	Lys	Tyr	Lys	Val	Ser	Tyr	Lys	Gly	
		115				120						125				
Pro	Gly	Pro	Gly	Ile	Lys	Phe	Ser	Ala	Glu	Ala	Leu	Arg	Cys	His	Leu	
		130				135						140				
Arg	Asp	His	Val	Asn	Val	Ser	Met	Val	Glu	Val	Thr	Asp	Phe	Pro	Phe	
		145				150				155				160		
Asn	Thr	Ser	Glu	Trp	Glu	Gly	Tyr	Leu	Pro	Lys	Glu	Ser	Ile	Arg	Thr	
				165				170						175		
Lys	Ala	Gly	Pro	Trp	Gly	Arg	Cys	Ala	Val	Val	Ser	Ser	Ala	Gly	Ser	
				180				185						190		
Leu	Lys	Ser	Ser	Gln	Leu	Gly	Arg	Glu	Ile	Asp	Asp	His	Asp	Ala	Val	
		195				200						205				
Leu	Arg	Phe	Asn	Gly	Ala	Pro	Thr	Ala	Asn	Phe	Gln	Gln	Asp	Val	Gly	
		210				215						220				
Thr	Lys	Thr	Thr	Ile	Arg	Leu	Met	Asn	Ser	Gln	Leu	Val	Thr	Thr	Glu	
		225				230				235				240		
Lys	Arg	Phe	Leu	Lys	Asp	Ser	Leu	Tyr	Asn	Glu	Gly	Ile	Leu	Ile	Val	
				245				250						255		
Trp	Asp	Pro	Ser	Val	Tyr	His	Ser	Asp	Ile	Pro	Lys	Trp	Tyr	Gln	Asn	
		260						265						270		
Pro	Asp	Tyr	Asn	Phe	Phe	Asn	Asn	Tyr	Lys	Thr	Tyr	Arg	Lys	Leu	His	
		275				280						285				
Pro	Asn	Gln	Pro	Phe	Tyr	Ile	Leu	Lys	Pro	Gln	Met	Pro	Trp	Glu	Leu	
		290				295				300						
Trp	Asp	Ile	Leu	Gln	Glu	Ile	Ser	Pro	Glu	Glu	Ile	Gln	Pro	Asn	Pro	
		305				310				315				320		
Pro	Ser	Ser	Gly	Met	Leu	Gly	Ile	Ile	Ile	Met	Met	Thr	Leu	Cys	Asp	
				325				330						335		
Gln	Val	Asp	Ile	Tyr	Glu	Phe	Leu	Pro	Ser	Lys	Arg	Lys	Thr	Asp	Val	
		340						345				350				
Cys	Tyr	Tyr	Tyr	Gln	Lys	Phe	Phe	Asp	Ser	Ala	Cys	Thr	Met	Gly	Ala	
		355				360						365				
Tyr	His	Pro	Leu	Leu	Tyr	Glu	Lys	Asn	Leu	Val	Lys	His	Leu	Asn	Gln	
		370				375				380						
Gly	Thr	Asp	Glu	Asp	Ile	Tyr	Leu	Leu	Gly	Lys	Ala	Thr	Leu	Pro	Gly	
		385				390				395				400		
Phe	Arg	Thr	Ile	His	Cys											
						405										

<210> SEQ ID NO 16

<211> LENGTH: 1218

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: nucleotides sequence of human a-2,6 sialic acid transferase

<400> SEQUENCE: 16

```

atgatccaca ccaacctgaa gaagaagttc agctgctgcg tgctggtggt cctgctgttc      60
gccgtgatct gcgtgtggaa ggagaagaag aaaggcagct actacgacag cttcaagctg      120
cagaccaagg agttocaggt gctgaagagc ctgggcaagc tggccatggg cagcgacagc      180
cagagcgtgt ccagctcttc caccaggat cccacagag gcagacagac cctgggcagc      240
ctgagaggcc tggccaaggc caagcccag gccagcttcc aggtgtggaa caaggacagc      300

```

-continued

```

agcagcaaga acctgatccc cagactgcag aagatctgga agaactacct gagcatgaac   360
aagtacaagg tgagctacaa aggacccgga cccggcatca agttcagcgc cgaggccctg   420
aggtgccacc tgagagacca cgtgaacgtg agcatgggtg aagtgaccga cttccccttc   480
aacaccagcg agtgggaagg ctacctgccc aaggagagca tcaggaccaa ggtgggcccc   540
tggggcagat gcgccgtggt gagcagcgct ggcagcctga agagctccca gctgggcaga   600
gagatcgacg accacgatgc cgtgctgagg ttcaatggcg ctcccaccgc caacttcag   660
caggacgtgg gcaccaagac cacaatccgg ctgatgaaca gccagctggt gacaaccgag   720
aagcgggttc tgaaggacag cctgtacaac gagggcatcc tgatcgtgtg ggatcccagc   780
gtgtaccaca gcgacatccc caagtgttac cagaatcccg actacaactt cttcaacaac   840
tacaagacct atagaaagct gcaccccaac cagcccttct acatcctgaa gccccagatg   900
ccctggggagc tgtgggacat cctgcaggag atcagccctg aagagatcca gcccacccct   960
ccctccagcg gcgatgctggg cattatcatc atgatgacct tgtgcgacca ggtggacatc  1020
tacgagttcc tgcccagcaa gagaaagacc gacgtgtgct actactatca gaagttcttc  1080
gacagcgctt gcaccatggg cgctaccac cccctgctgt acgagaagaa cctggtgaag  1140
cacctgaacc agggcaccga cgaggacatc tacctgctgg gcaaagccac cctgcccggc  1200
ttcagaacca tccactgc                                     1218

```

```

<210> SEQ ID NO 17
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgD hinge variant
<220> FEATURE:
<221> NAME/KEY: Xaa is Lys or Gly
<222> LOCATION: (12)..(12)
<220> FEATURE:
<221> NAME/KEY: Xaa is Glu, Gly, or Ser
<222> LOCATION: (13)..(13)

```

```

<400> SEQUENCE: 17

```

```

Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa Xaa Lys Glu Lys
1             5             10            15

Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro
                20            25            30

```

```

<210> SEQ ID NO 18
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgD hinge variant
<220> FEATURE:
<221> NAME/KEY: Xaa is Lys or Gly
<222> LOCATION: (31)..(31)
<220> FEATURE:
<221> NAME/KEY: Xaa is Glu, Gly, or Ser
<222> LOCATION: (32)..(32)

```

```

<400> SEQUENCE: 18

```

```

Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr Thr Ala Pro
1             5             10            15

Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa Xaa
                20            25            30

Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys
                35            40            45

```

-continued

Pro

<210> SEQ ID NO 19
 <211> LENGTH: 30
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgD hinge

<400> SEQUENCE: 19

Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Lys Glu Lys Glu Lys
 1 5 10 15
 Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro
 20 25 30

<210> SEQ ID NO 20
 <211> LENGTH: 425
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FceRIa ECD-hinge-Fc1

<400> SEQUENCE: 20

Val Pro Gln Lys Pro Lys Val Ser Leu Asn Pro Pro Trp Asn Arg Ile
 1 5 10 15
 Phe Lys Gly Glu Asn Val Thr Leu Thr Cys Asn Gly Asn Asn Phe Phe
 20 25 30
 Glu Val Ser Ser Thr Lys Trp Phe His Asn Gly Ser Leu Ser Glu Glu
 35 40 45
 Thr Asn Ser Ser Leu Asn Ile Val Asn Ala Lys Phe Glu Asp Ser Gly
 50 55 60
 Glu Tyr Lys Cys Gln His Gln Gln Val Asn Glu Ser Glu Pro Val Tyr
 65 70 75 80
 Leu Glu Val Phe Ser Asp Trp Leu Leu Leu Gln Ala Ser Ala Glu Val
 85 90 95
 Val Met Glu Gly Gln Pro Leu Phe Leu Arg Cys His Gly Trp Arg Asn
 100 105 110
 Trp Asp Val Tyr Lys Val Ile Tyr Tyr Lys Asp Gly Glu Ala Leu Lys
 115 120 125
 Tyr Trp Tyr Glu Asn His Asn Ile Ser Ile Thr Asn Ala Thr Val Glu
 130 135 140
 Asp Ser Gly Thr Tyr Tyr Cys Thr Gly Lys Val Trp Gln Leu Asp Tyr
 145 150 155 160
 Glu Ser Glu Pro Leu Asn Ile Thr Val Ile Lys Ala Pro Arg Glu Lys
 165 170 175
 Tyr Trp Leu Gln Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys
 180 185 190
 Glu Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu
 195 200 205
 Cys Pro Ser His Thr Gln Pro Leu Gly Val Phe Leu Phe Pro Pro Lys
 210 215 220
 Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
 225 230 235 240
 Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr
 245 250 255
 Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
 260 265 270

-continued

Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His
 275 280 285

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
 290 295 300

Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln
 305 310 315 320

Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu Met
 325 330 335

Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
 340 345 350

Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn
 355 360 365

Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu
 370 375 380

Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn Val
 385 390 395 400

Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln
 405 410 415

Lys Ser Leu Ser Leu Ser Leu Gly Lys
 420 425

<210> SEQ ID NO 21
 <211> LENGTH: 425
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FceRIa ECD-hinge-Fc2

<400> SEQUENCE: 21

Val Pro Gln Lys Pro Lys Val Ser Leu Asn Pro Pro Trp Asn Arg Ile
 1 5 10 15

Phe Lys Gly Glu Asn Val Thr Leu Thr Cys Asn Gly Asn Asn Phe Phe
 20 25 30

Glu Val Ser Ser Thr Lys Trp Phe His Asn Gly Ser Leu Ser Glu Glu
 35 40 45

Thr Asn Ser Ser Leu Asn Ile Val Asn Ala Lys Phe Glu Asp Ser Gly
 50 55 60

Glu Tyr Lys Cys Gln His Gln Gln Val Asn Glu Ser Glu Pro Val Tyr
 65 70 75 80

Leu Glu Val Phe Ser Asp Trp Leu Leu Leu Gln Ala Ser Ala Glu Val
 85 90 95

Val Met Glu Gly Gln Pro Leu Phe Leu Arg Cys His Gly Trp Arg Asn
 100 105 110

Trp Asp Val Tyr Lys Val Ile Tyr Tyr Lys Asp Gly Glu Ala Leu Lys
 115 120 125

Tyr Trp Tyr Glu Asn His Asn Ile Ser Ile Thr Asn Ala Thr Val Glu
 130 135 140

Asp Ser Gly Thr Tyr Tyr Cys Thr Gly Lys Val Trp Gln Leu Asp Tyr
 145 150 155 160

Glu Ser Glu Pro Leu Asn Ile Thr Val Ile Lys Ala Pro Arg Glu Lys
 165 170 175

Tyr Trp Leu Gln Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Gly
 180 185 190

Ser Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu
 195 200 205

-continued

Cys Pro Ser His Thr Gln Pro Leu Gly Val Phe Leu Phe Pro Pro Lys
 210 215 220
 Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
 225 230 235 240
 Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr
 245 250 255
 Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
 260 265 270
 Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His
 275 280 285
 Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
 290 295 300
 Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln
 305 310 315 320
 Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu Met
 325 330 335
 Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
 340 345 350
 Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn
 355 360 365
 Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu
 370 375 380
 Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn Val
 385 390 395 400
 Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln
 405 410 415
 Lys Ser Leu Ser Leu Ser Leu Gly Lys
 420 425

<210> SEQ ID NO 22
 <211> LENGTH: 444
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FcεRIα ECD-hinge-Fc3

<400> SEQUENCE: 22

Val Pro Gln Lys Pro Lys Val Ser Leu Asn Pro Pro Trp Asn Arg Ile
 1 5 10 15
 Phe Lys Gly Glu Asn Val Thr Leu Thr Cys Asn Gly Asn Asn Phe Phe
 20 25 30
 Glu Val Ser Ser Thr Lys Trp Phe His Asn Gly Ser Leu Ser Glu Glu
 35 40 45
 Thr Asn Ser Ser Leu Asn Ile Val Asn Ala Lys Phe Glu Asp Ser Gly
 50 55 60
 Glu Tyr Lys Cys Gln His Gln Gln Val Asn Glu Ser Glu Pro Val Tyr
 65 70 75 80
 Leu Glu Val Phe Ser Asp Trp Leu Leu Leu Gln Ala Ser Ala Glu Val
 85 90 95
 Val Met Glu Gly Gln Pro Leu Phe Leu Arg Cys His Gly Trp Arg Asn
 100 105 110
 Trp Asp Val Tyr Lys Val Ile Tyr Tyr Lys Asp Gly Glu Ala Leu Lys
 115 120 125
 Tyr Trp Tyr Glu Asn His Asn Ile Ser Ile Thr Asn Ala Thr Val Glu
 130 135 140

6. An isolated host cell comprising the expression vector according to claim 5.

7. A method for producing the polypeptide dimer of claim 1, comprising: a step of culturing a cell, into which (a) a polynucleotide encoding a monomer of the polypeptide dimer and (b) a sialic acid transferase gene is introduced, under a condition for expression of the polynucleotide, wherein the monomer contains an extracellular domain of an alpha subunit of an IgE Fc receptor (FcεRIα-ECD) consisting of the amino acid sequence of SEQ ID NO: 1; wherein the monomer comprises a modified Fc region consisting of the amino acid sequence of SEQ ID NO: 2; wherein the N-terminus of the modified Fc region is linked via a hinge to the C-terminus of the FcεRIα-ECD to have a structure of N'-FcεRIα-ECD-hinge-modified Fc region-C'; and wherein the hinge comprises the amino acid sequence of SEQ ID NO: 3; and a step of recovering the polypeptide dimer.

8. The polypeptide dimer obtained by the method of claim 7.

9. A pharmaceutical composition comprising, as an active ingredient, the polypeptide dimer according to claim 8.

10. A food composition comprising, as an active ingredient, the polypeptide dimer according to claim 8.

11. A method for treating a food allergy in a subject in need thereof, comprising: a step of administering to the subject an effective amount of the polypeptide dimer according to claim 1.

12. A method for treating a food allergy in a subject in need thereof, comprising: a step of administering to the subject an effective amount of the polypeptide dimer according to claim 8.

* * * * *